

MSc in Information Technology

Specialized in Enterprise Applications Development

LabLog: Design and Implementation of an Offline-First Digital Practical Laboratory Book

Module:	Enterprise Mobility	**Student ID:**	MS26928210
Code:	SE5070	**Name:**	Peiris D. T. M.
Assignment:	Assignment 1	**Semester No:**	1
		Date:	2026-10-18

Contents

Abstract / Executive Summary 4

1. Introduction and Project Domain 5

    1.1 Project area calculation 5

    1.2 Report scope and evidence 5

2. Problem Definition, Objectives and Features 6

    2.1 Problem definition 6

    2.2 Proposed solution and objectives 6

    2.3 Project rationale 6

    2.4 Feature list 7

    2.5 Distinctive design characteristics 8

3. Application Architecture and Design 9

    3.1 Layers and responsibilities 9

    3.2 Major design decisions 10

    3.3 Data flow between the layers 12

    3.4 How the source is organised, and why 14

    3.5 Backend API 15

4. Offline Access and Synchronisation 17

5. Constraint Algorithms and Design Principles 20

    5.1 Model 20

    5.2 The unlock algorithm 20

    5.3 Graph validation (Kahn's algorithm) 21

    5.4 Principles applied 21

    5.5 Designing for a constrained device 22

6. Native Hardware and Platform Integration 24

7. User Interface and User Experience 26

    7.1 Navigation 26

7.2 Screen by screen 26
7.3 Interface design rationale 30
8. Custom Component Design 33
8.1 ExperimentChecklist (the main custom component) 33
8.2 Other custom components 35
9. Decision records, failure logs and AI use 36
9.1 How the process was recorded 36
9.2 Failure log 36
9.3 AI-Assisted Development 43
10. Testing and Evaluation 44
11. Limitations and Future Work 47
11.1 Operational and synchronisation limits 47
11.2 Security and evidence limits 47
11.3 Functional scope and evaluation limits 47
12. Conclusion 49
References 50
Appendices 51
A. Running the project 51
B. Environment 51
C. Size 51
D. Academic integrity 51
E. Assignment Requirements Traceability 52
F. Report Preparation Note 52
G. AI-Assisted Development: Development Prompt Register 53

Abstract / Executive Summary

LabLog is a native iOS application for recording university laboratory practicals. It allows students to record their progress and supporting evidence when an internet connection is unreliable. The project falls within the Edtech domain, as determined by the index-number calculation in the SE5070 assignment brief [1]. The application uses SwiftUI for its interface, SwiftData for local storage, a custom Express and TypeScript backend, and Supabase for authentication, database storage and photographs.

Each experiment consists of steps linked by prerequisites. Some steps also require a photograph or a minimum waiting period. A shared constraint engine checks these conditions before allowing a step to be completed. The application saves changes on the phone first and uploads them later, using an outbox, stable record identifiers, retries and revision checks. Students can attach photographs, add observations and request a location stamp when submitting. Lecturers can create experiments, enrol students and review submitted records.

The evaluation draws on automated tests, development records and checks carried out on the test device [2] to [5]. The automated suites pass on the simulator and on a physical iPhone against the deployed backend. The main limitations, discussed in Section 11, are single-author synchronisation, lecturer features that require a connection, and incomplete server-side enforcement of procedural rules. The report explains these limits alongside the design decisions, failures and AI assistance recorded during development.

1. Introduction and Project Domain

1.1 Project area calculation

The assignment allocates a project domain using the digit sum of the student index number, followed by division modulo five [1]. Table 1 presents the calculation for this project.

Table 1. Index-number calculation and allocated project domain.

Calculation stage	Value or result
Student index number	MS26928210
Numerical characters	2, 6, 9, 2, 8, 2, 1, 0
Digit sum	2 + 6 + 9 + 2 + 8 + 2 + 1 + 0 = 30
Modulus calculation	30 mod 5 = 0
Allocated project domain	0: Educational Technology (Edtech)

LabLog addresses the allocated Edtech domain by supporting the structured recording and review of university practical work.

1.2 Report scope and evidence

This report explains how the mobile application and backend work together, focusing on offline access, procedural constraints, native hardware and custom components. The discussion draws on the source repositories, architecture decisions and failure logs [2] to [5]. The evaluation covers the application's functional behaviour, and Section 11 sets out its limits.

2. Problem Definition, Objectives and Features

2.1 Problem definition

LabLog was designed for practical sessions in which students follow a procedure and record observations and evidence. Paper records and general-purpose documents leave students to track step order and evidence requirements themselves. A mobile application can guide that process, but it must also allow work to continue when the network is unavailable. Electronic notebooks already replace paper in research laboratories [7], but they are built for open-ended research notes rather than for guiding a student through a set procedure.

2.2 Proposed solution and objectives

LabLog organises each experiment as a checklist with rules. A student can complete a step only after its prerequisites, waiting period and evidence requirements have been met. The resulting record contains completion times, notes, photographs and observations. The application requests a location stamp during submission, but a missing location does not prevent the student from submitting the record.

The main objectives are to keep practical work available offline, apply the same step rules throughout the app, use the camera and location services, and provide a reusable checklist component. The backend handles sign-in, catalogue access, synchronisation and lecturer operations. Students sign in before using the application and can then continue local work offline. Lecturers have separate screens for creating experiments, choosing students and reviewing submissions.

2.3 Project rationale

Practical work is where students learn to carry out a procedure and to record what they observe, and it is a central part of science education [6]. The record of that work is also what a lecturer assesses. On paper, a skipped step, a step done out of order or a missing observation is usually discovered only after the session, when it can no longer be corrected. Electronic laboratory notebooks show that digital records can replace paper [7], but they serve researchers keeping open-ended notes. LabLog targets the teaching case: it guides the student through a fixed procedure while it records the evidence.

Three properties make the idea worthwhile in this domain. First, the procedure itself is modelled: an experiment is a dependency graph whose order is checked with a standard

topological-sorting method [9], so the app can show the next valid step, explain why a step is locked, and refuse an impossible procedure before it reaches a student. Second, the app is local-first [8]: laboratories are often in basements or shielded rooms with poor connectivity, so every action is saved on the phone and delivered later through an outbox [11], and a dropped connection never costs the student their work. Third, the record carries its context: completion times, a photograph on the step that needs it and a location stamp give the lecturer consistent, comparable records to review in one place, rather than a set of loose files.

The approach also extends beyond one module. Experiments are data rather than code, so a lecturer can author a new procedure with its rules in the app (ADR-029), and the same checklist, constraint engine and synchronisation serve chemistry, physics, biology or engineering practicals alike.

2.4 Feature list

Feature	What it does	Brief requirement
Lab and experiment catalog	Browse labs and experiments; uses a bundled catalogue after initial authentication, refreshed from the API	Backend, offline
Constrained step checklist	Steps unlock by precedence; photo and timer guards; explains *why* a step is locked	Constraint design
Step guide	How-to sub-steps, safety note, expected result; jump to blocking steps	UX
Photo evidence	Camera capture per step, stored on the device, uploaded later	Native hardware (camera)
Location stamp	One-shot GPS fix on submit, shown on a map receipt	Native hardware (GPS)
Live timers	Countdown dial in the app, Dynamic Island and Lock Screen Live Activity, "ready" notification, *Mark done* from the island	Native platform features
Offline outbox and sync	Local-first storage; automatic sync on reconnect, with retry and backoff	Offline + sync
Review and submit	Summary of what will be submitted; records become read-only	UX

Feature	What it does	Brief requirement
Records and sync status	All records with progress and sync state; connection, account and outbox status	UX
Lecturer review	A Review tab for lecturers: submissions per experiment, with steps and times, photos, observation, step notes and map; backed by a lecturer-only API	Backend, UX
Editing a draft	Retake a step's photo (remove it until the step is done), add a note to any completed step, or discard the whole draft; every change syncs, deletes included	Offline + sync, constraint design
Server address	The submitted build uses the hosted backend at a fixed address, shown read-only in Sync. A local development build can instead point at the Mac through Sync → Server, with a connection test (ADR-021)	Backend, UX
Accounts	Sign in first, then work offline; sign-out clears the phone and signing back in restores the student's records with their photos	Offline + sync, backend
Lecturer lab management	Lecturers create labs and experiments (steps with photo, wait and prerequisite rules, checked by the constraint engine in the app and with Kahn's algorithm on the server) and choose which students can do each lab	Constraint design, backend, UX

2.5 Distinctive design characteristics

LabLog represents an experiment as a graph of steps and conditions. The checklist, step guide, Live Activity action and submission check all use the same constraint engine, which keeps their behaviour consistent. The backend separately checks identity and record ownership. Client rules guide students through a procedure, while server controls decide which data a user may access or change.

3. Application Architecture and Design

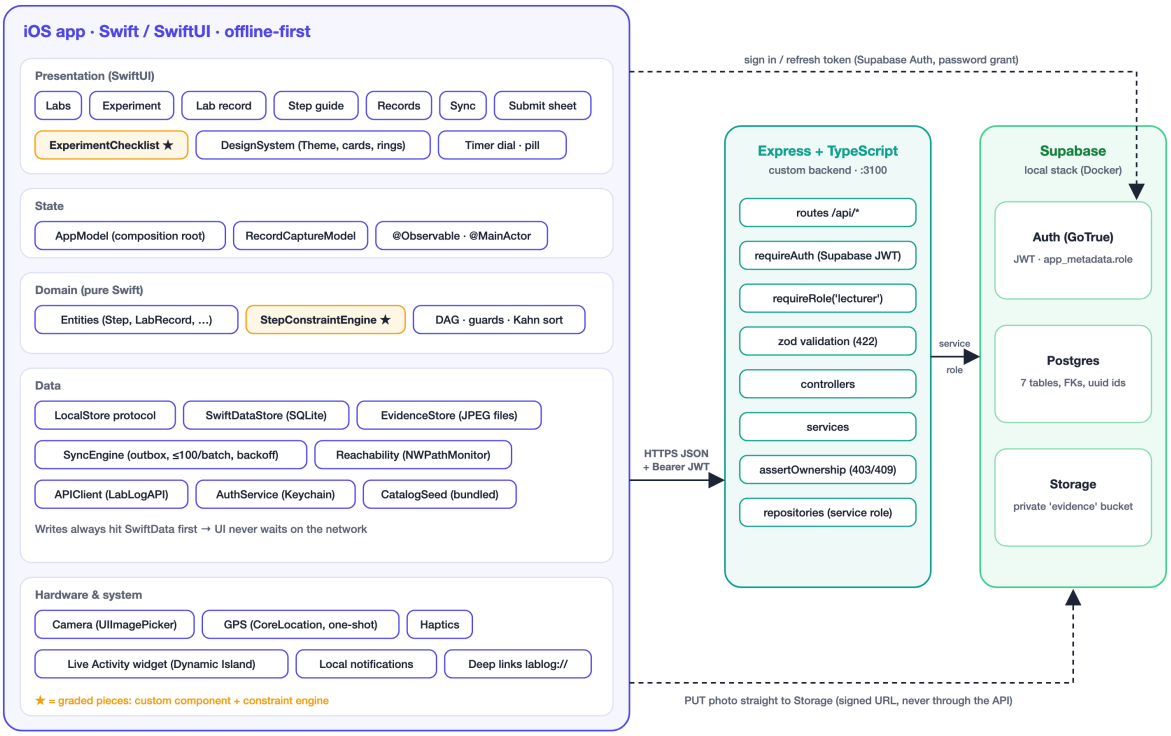


Figure 1. Layers of the iOS app, the custom API, and Supabase. ★ marks the custom component and the constraint engine.

3.1 Layers and responsibilities

Layer	Contents	Responsibility
Presentation	SwiftUI screens, ExperimentChecklist, design system	Render state and turn taps into intents. No business rules.
State	AppModel (composition root), RecordCaptureModel (@Observable, @MainActor)	Hold screen state, call the domain and data layers, expose derived values (progress, next step, can-submit)
Domain	Entities, StepConstraintEngine	Pure Swift rules with no frameworks. The single source of truth for what a student may do.
Data	LocalStore → SwiftDataStore, EvidenceStore, SyncEngine, Reachability, APIClient, AuthService	Persistence, the outbox, networking and authentication

Layer	Contents	Responsibility
Hardware and system	Camera, CoreLocation, haptics, the Live Activity widget, notifications, deep links	Device capabilities, each behind a small wrapper
API (Express)	routes → middleware → controllers → services → repositories	Authentication, validation, ownership rules, data access
Supabase	Auth (JWT), Postgres, Storage	Identity, relational data, photo files

3.2 Major design decisions

The project records thirty architecture decisions in lablog-ios/docs/decision-records.md [4]. Each entry explains the context, the decision and its consequences. The table below summarises the decisions that had the greatest effect on the application.

ADR	Decision	Why
001	Native iOS with SwiftUI	First-class camera, location, Live Activities and SwiftData; a single demo platform
002	A custom Express API in front of Supabase, not direct database access from the app	A real service layer for validation, authorisation and business rules; the service-role key never ships in the app
003	The local store is the source of truth; the network is a background concern	The app must work fully offline in the lab
004	Client-generated UUIDs and idempotent upserts	Retrying after a dropped connection never creates duplicates
005	Photos uploaded through signed Storage URLs, not inside the JSON	Small sync payloads; photos never pass through the API
006 / 012	All step rules in one pure engine; step graphs validated with Kahn's algorithm	One place to reason about correctness; impossible experiments are rejected
008	Catalog bundled with the app, using the same UUIDs as the server seed	Catalogue available after initial sign-in, with stable identifiers linking records to server rows
009	A per-record revision counter guards in-flight edits	An edit made during a sync is never lost

ADR	Decision	Why
013	Design system and a timeline checklist	Consistent visual language; the next step is always the most visible thing
015	Timed steps as a Live Activity (Dynamic Island)	Students can leave the app during a wait
018	A local Supabase stack for end-to-end testing	Proves the full path on real Postgres, Auth and Storage
019	Lecturer role in app_metadata; ownership checks during sync	Closes record reading and record hijacking holes
020	In-app lecturer review, fetched live; the role only changes what the UI shows, and the API enforces it	Both roles demonstrable end to end; no trust placed in the client
021	The server address can change at runtime; the API points storage URLs at the host the app called	The demo works on any network, including a phone hotspot, with no rebuild
022	Discarding a synced draft sends a tombstone; the server deletes drafts only	Deletes are offline-first like every other write; submitted evidence is immutable
023	Photo retake (remove only until the step is done) and step notes; the payload carries the full photo set	Evidence can be corrected without breaking the photo guard or leaving stale files (ids compared in lower case since failure 018)
024	A quiet sync icon instead of an "Online" pill; a tap explains the state in one sentence	Being online is normal, so it is not news; the unusual states are explicit
025	Host the API on Vercel and the database on Supabase, every table locked by RLS; iPhone builds use it (and, since ADR-030, the simulator and tests)	No dependence on the Mac's address; the anon key in the app cannot read or write tables directly
026	Signing out clears the phone's records (warning first if any have not uploaded)	One person's work is never shown to, or uploaded as, the next person on a shared phone
027	Sign in first, then work offline, like a messaging app (supersedes the optional sign-in of ADR-010)	Every record has a known owner from the start; offline work after the first sign-in is unchanged
028	Restore the student's records from the server after signing in (GET /api/records/mine), local copies winning	Sign-out can clear the phone without the student losing their history

ADR	Decision	Why
029	Lecturers create labs and experiments (with step rules) and enrol students; tabs follow the role	The catalog is authored in the app; constraint checks run when an experiment is written, not only when it is done
030	Test against the hosted backend (Vercel + Supabase) with test-only accounts, cleaned up after every run; Docker optional	The tests check exactly what the phone uses, and do not depend on a local Docker stack that kept quitting (025, 029)

3.3 Data flow between the layers

Data flows through the application in one direction. Views send user actions to observable models, the models call the domain and data layers, and changes in stored state update the interface. This keeps network requests and business rules separate from the views.

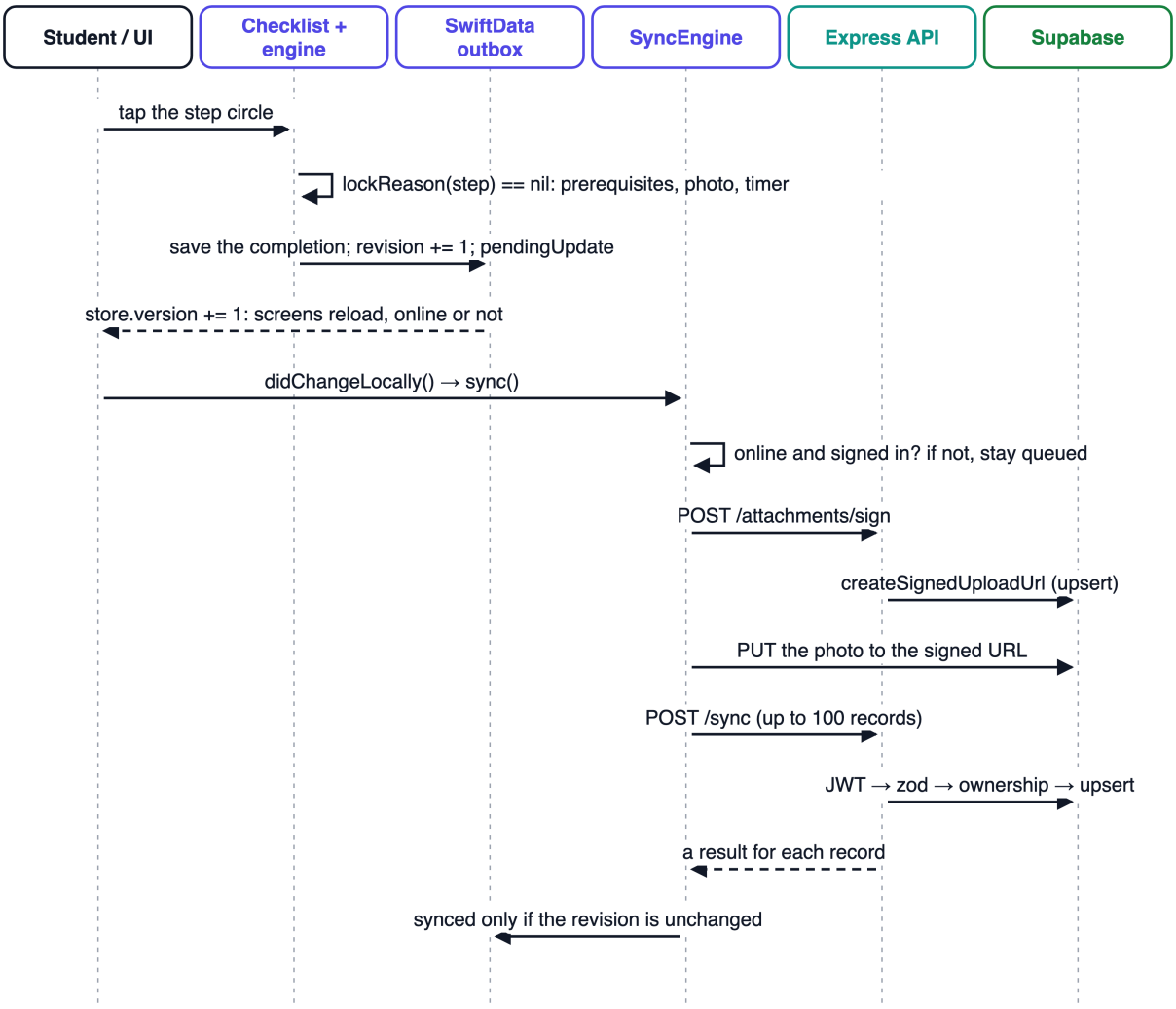


Figure 2. What happens when a student completes a step: a local write first, then the outbox drains to the server.

Write path (Figure 2).

1. The student taps a step's circle in ExperimentChecklist.
2. The component asks StepConstraintEngine.lockReason(for:context:). If the result is nil, it creates a StepCompletion.
3. The completion goes to SwiftDataStore. That increments the record's revision, marks it pendingUpdate, and bumps store.version.
4. Every screen that watches store.version reloads immediately, with or without a network.
5. AppModel.didChangeLocally() calls SyncEngine.sync(). If the phone is offline or signed out, the record simply stays queued; NWPathMonitor triggers the sync again on reconnect.
6. The engine uploads pending photos: POST /attachments/sign, then a PUT straight to Storage.

7. It then sends the records in batches of up to 100 (POST /sync). On the server, requireAuth checks the JWT, zod validates the payload, and assertOwnership blocks hijacking before the upserts run.

8. The per-item results come back, and applySyncResult marks a record synced only if its revision has not changed in the meantime.

Read path. Screens read only from SwiftData, through the LocalStore protocol, so the UI never waits on the network. When the phone is online and signed in, SyncEngine.refreshCatalog() pulls GET /labs, /labs/:id/experiments and /experiments/:id (the experiments concurrently), validates each step graph, updates the local catalog and drops labs the server no longer lists for this user (ADR-029). After each sync, pullRecords() fetches GET /records/mine and adds any of the student's records the phone lacks, with their photos (ADR-028).

Patterns used (from the lectures):

- **MVVM with observable models**: views bind to @Observable models.

- **Repository pattern** [10]: the LocalStore protocol on the app; repositories on the server.

- **Dependency injection**: the AppModel composition root builds every service once and passes it to views through the environment.

- **Protocol abstraction for testability**: LabLogAPI lets the sync engine be tested against a fake backend.

- **Layered API**: routes → controllers → services → repositories.

3.4 How the source is organised, and why

```
lablog-ios/
  LabLog/
    App/          entry point, AppModel (composition root), config, tabs
    DesignSystem/  Theme tokens + shared components (cards, rings, chips)
    Features/      one folder per screen: LabList, ExperimentList,
                   ExperimentDetail, RecordCapture (capture, step guide,
                   submit sheet), Records, SyncStatus, Auth,
                   Review (lecturer), Manage (lecturer labs)
    Components/    ExperimentChecklist ★ (component, row, style), badges
    Domain/        Entities, Constraints/StepConstraintEngine ★ (pure Swift)
    Data/          Local (SwiftData store, catalog seed, evidence files)
                   Remote (API client, DTOs, Supabase auth, Keychain)
                   Sync (outbox, reachability)
    Hardware/      camera and location wrappers
    LiveActivity/  timer coordinator (activity + notifications)
  Shared/          compiled into app and widget (activity data, intent)
  LabLogWidgets/   widget extension: Dynamic Island + Lock Screen
  LabLogTests/     63 unit tests
```

```
  LabLogUITests/   6 end-to-end UI tests
  docs/            decision records, failure log, screenshots

lablog-backend/
  src/       routes → controllers → services → repositories,
             middleware (auth, roles, zod), dto
  db/        schema.sql, seed.sql, migrations/
  supabase/  local stack config
  test/      e2e.test.ts (20 tests)
  scripts/   local setup, demo users
```

Why this structure:

- **Feature folders** keep a screen's view and model together, so a change touches one place.
- **The domain layer has no framework imports**, so the most important rules can be unit-tested in milliseconds and reused by the app, the guide and the Live Activity intent.
- **The data layer sits behind protocols** (LocalStore, LabLogAPI), so SwiftData or the network can be swapped or faked.
- **`Shared/` exists only because the widget extension is a separate binary.** It holds the minimum both need.
- **The backend mirrors the same separation** (transport → logic → data), so each layer is testable and the business rules sit in services, not controllers.

3.5 Backend API

Method	Path	Access	Purpose
GET	/api/health	public	Liveness check
GET	/api/labs · /api/labs/:id/experiments · /api/experiments/:id	signed in	Catalog, including the step graph and guides. Students see the open course labs and the labs they are enrolled in; any other lab is 403
POST	/api/attachments/sign	student	Signed upload URL for a photo (deterministic path, upsert)
POST	/api/sync	student	Batch upsert of records and their children; result per record. deleted: true deletes a draft (tombstone); photos missing from a record's set are removed

Method	Path	Access	Purpose
GET	/api/records/mine	student	The student's own records (drafts and submitted) with signed photo URLs, to restore a phone after sign-in (ADR-028)
GET	/api/records?experimentId=	**lecturer**	Submitted records with student email and 1-hour signed photo URLs
GET · POST · PUT	/api/manage/labs · /api/manage/labs/:labId/experiments · /api/manage/experiments/:id · /api/manage/labs/:labId/students · /api/manage/students	**lecturer**	Create labs and experiments (step graph checked with Kahn's algorithm, 422 on a loop), edit an experiment in place (steps keep their ids; 409 if a removed step has student work), and choose which students can do each lab (ADR-029)

Data model (Postgres):

- Catalog: labs 1- *`experiments` 1-* steps. Each step has prerequisite_step_ids uuid[], requires_photo, min_duration_seconds and its guide content. labs.created_by is null for the open course labs; lab_enrolments (lab, student) decides who sees a lecturer's lab.
- Records: lab_records (owned by auth.users) 1-* step_completions, observations and attachments.
- All record ids are client-generated UUIDs.
- On a local development stack, storage URLs that the API hands to the app (signed uploads and lecturer photo links) use whatever host the app called, so the Mac's address can change without touching the backend (ADR-021). The hosted project has one fixed address, and its tables are protected by row-level security [14].

4. Offline Access and Synchronisation

LabLog follows a local-first design [8]: it saves practical work locally before sending it to the server through an outbox [11]. Screens read from the local store, so students can keep working when the connection drops. Offline use requires an initial sign-in and a catalogue already stored on the phone. The bundled catalogue does not remove the sign-in requirement.

Technique	**How it is implemented**
Bundled offline catalogue after initial authentication	CatalogSeed bundles 2 labs, 3 experiments and 13 steps with the same UUIDs as seed.sql, versioned so updates reach existing installs
Local persistence	SwiftData models mirror the domain structs; every action writes locally first
Outbox	Each record carries a local syncState (pendingCreate, pendingUpdate, synced, failed); every non-synced record is in the outbox
No lost edits	A revision counter: a sync result marks a record synced only if no edit happened while the request was in flight
Safe retries	Client UUIDs and upserts by id; the photo path is deterministic and uploads overwrite (upsert: true)
Photos offline	EvidenceStore keeps downscaled JPEGs (1600 px, quality 0.7) in Application Support, by relative file name, until uploaded
Reconnect detection	NWPathMonitor: the offline → online transition triggers a catalog refresh, then a sync
Retry policy	A failed sync retries with exponential backoff (2, 4, 8, 16, 32 s); reconnecting or *Sync now* retries immediately
Efficient sync	Photos upload first; records go in batches of up to 100; the server reports success or failure per record, so one bad record does not block the rest
Sign in once, then offline	The app opens on sign-in (ADR-027); after that the session lives in the Keychain, so the app works with no connection, and the token refreshes a minute before expiry
Deletes offline too	A discarded draft the server already has becomes a **tombstone**: hidden everywhere, kept in the outbox, sent as deleted: true, purged once the server confirms. A draft never sent is just deleted (ADR-022)

Technique	How it is implemented
Edits as full state	A record's payload lists its complete photo set, so a retaken or removed photo is dropped on the server too (rows and files)
Any network	The app, the simulator and the tests use the hosted backend at a fixed HTTPS address, so they work on any network without configuration (ADR-025, ADR-030). Only a local development build has an editable server address (ADR-021)
Sign-out	Returns to the sign-in screen and clears the phone's records, photos and timers; if any have not uploaded, the student is warned first (ADR-026)
Restore after sign-in	After each sync the app fetches the student's own records and adds the ones it lacks, with their photos; local records always win (ADR-028)

What the student sees: a sync icon on every main screen that stays quiet when everything is synced and becomes specific when it is not (an upload cloud with the number of waiting records, *Offline*, or a warning). Tapping it explains the status in one sentence, with the one action that helps: *Sync now*, or *Sign in* if the session has ended while the app is open, which opens the sign-in sheet right there (ADR-024). There is also a sync state on every record, and a Sync screen that leads with one clear status ("Working offline", "3 records ready to sync", "Everything is synced"). Its main button is whatever unblocks sync next (*Sign in* or *Sync now*).

Evidence:

- **Unit tests:** 18 store tests and 9 sync-engine tests cover outbox rules, the revision guard, photo-before-record ordering, per-item failures, tombstones, photo removal and notes, restoring a student's records after sign-in, clearing them on sign-out, and the offline and signed-out behaviour. Four more check what the sync icon shows in each state.
- **Backend tests:** 20 end-to-end tests against real Postgres, Auth and Storage, run against the deployed API on Vercel (ADR-030), including a tombstone retried twice, a retake replacing the stored photo, ids in the app's upper-case form keeping their photo (failure 018), and, on a local stack, storage URLs following the caller's host.
- **UI test:** it signs in, completes steps with a photo, and waits for "Everything is synced". The record, both step completions and the photo were then confirmed in the database. It then signs out (Records is empty), signs back in and finds the record restored (Figure 3).

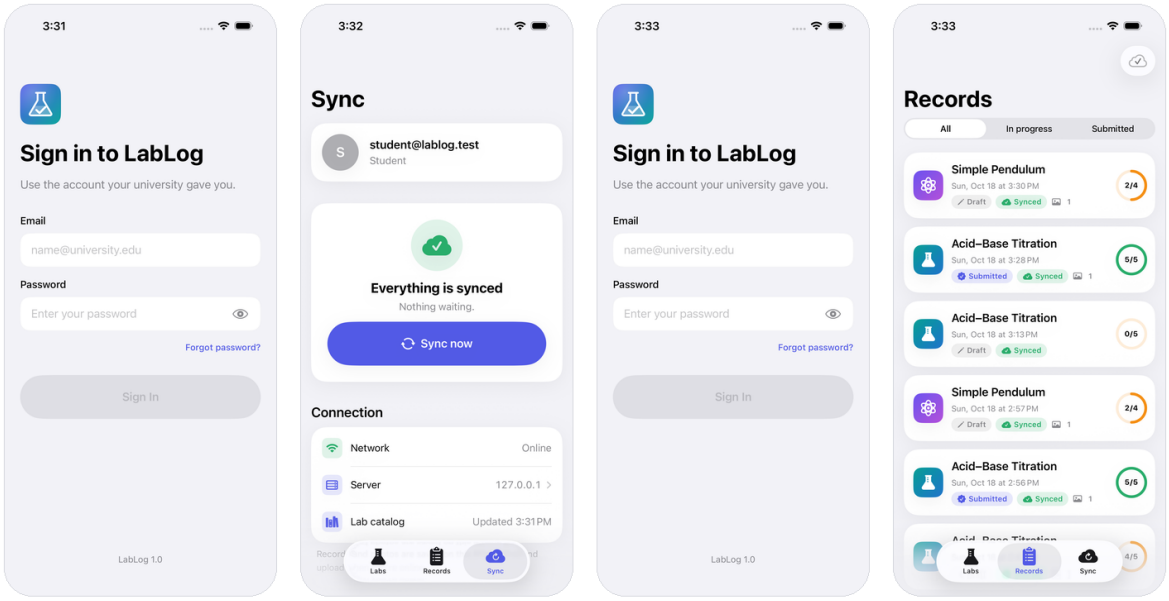


Figure 3. The app opens on sign-in (Supabase Auth); after work, the outbox drained to the server; signing out returns to sign-in and clears the phone; signing in again restores the student's records from the server (ADR-028).

5. Constraint Algorithms and Design Principles

5.1 Model

An experiment is a **directed acyclic graph of steps**. Each step lists its prerequisites (prerequisiteStepIds) and may carry **guard conditions**:

- requiresPhoto: a photo must be attached to this step.

- minDurationSeconds: a minimum wait, measured from the latest prerequisite's completion, or from the record's start when the step has no prerequisites (when the step unlocked).

The seeded *Acid-Base Titration* experiment is a real branching graph. Steps 1 and 2 can be done in either order, and Step 4 needs **both** Step 1 and Step 3.

5.2 The unlock algorithm

```
lockReason(step, ctx):
  missing ← step.prerequisites − ctx.completed
  if missing ≠ ∅                    → prerequisitesIncomplete(missing)   // precedence
  if step.minDuration ≠ nil:
    anchor ← max(completionTime[p] for p in step.prerequisites)          // latest prereq
          ?? recordStartedAt                                             // else: its start
    remaining ← ceil(minDuration − (now − anchor))
    if remaining > 0                → timerPending(remaining)          // time guard
  if step.requiresPhoto ∧ step ∉ ctx.photos → needsPhoto                 // evidence guard
  return nil                                                             // unlocked
```

The engine checks prerequisites first, followed by the waiting period and photographic evidence. This order helps the student understand what needs attention next. A timer starts from the latest prerequisite completion, so an earlier parallel step cannot shorten the wait. If a step has no prerequisites, the record creation time is used. Failure 028 showed why this matters: using the current time on every check restarted the wait indefinitely. The engine takes state and time as inputs, allowing the same scenario to be tested repeatedly with the same result.

The engine also provides:

- reachableSteps(…): what the student can work on next. It drives "Next: Step 3 · …".

- isComplete(…): whether every step is done. It gates submission.

- message(for:): a human-readable reason, such as "Complete Step 1, Step 3 first" or "Wait 2:30".

5.3 Graph validation (Kahn's algorithm)

An experiment cannot be completed if a prerequisite points to a missing step or if its steps depend on one another in a cycle. The application checks the graph before accepting or displaying the experiment to prevent these situations.

The validate(_:) function checks for self-dependencies, unknown prerequisites and cycles. It uses Kahn's algorithm [9] to place steps in an order that respects their dependencies. Any steps left over depend on a cycle or form part of one. Catalogue refresh skips invalid experiments, and the server checks the graph again when a lecturer creates or edits an experiment.

The experiment editor offers earlier steps as prerequisites and checks the complete draft before enabling Save. The server also validates the graph and returns HTTP 422 for invalid definitions. When an experiment is edited, existing step identifiers are kept so that recorded completions remain linked to the correct steps. Removing a step that already has student work is rejected with HTTP 409.

The current implementation scans the step collection repeatedly, with an upper-bound cost of $O(V \cdot (V+E))$, where V is the number of steps and E is the number of prerequisite relationships. This is acceptable for the small procedures used in the project. An adjacency-list implementation could reduce topological sorting to $O(V+E)$, with an additional ordering cost if a fixed priority among available steps is required.

5.4 Principles applied

1. **Precedence constraints.** A topological order is enforced, so Step 4 cannot be reached before Steps 1 and 3.
2. **Guard conditions.** Evidence and time preconditions are first-class, not ad-hoc ifs.
3. **Invariant centralisation (single source of truth).** The checklist, the step guide, the Dynamic Island's *Mark done* intent and the submit gate all ask the same engine, so no screen can disagree with another.
4. **Fail-safe, explanatory UI.** Locked steps stay tappable and say *why* they are locked (with a warning haptic); the guide lists the blockers and jumps to them.
5. **Defence in depth.** The server re-validates every payload with zod (422 on bad input), checks ownership (403 or 409), and limits batch size.

6. **Validate before use.** Graph validation rejects an impossible configuration before it reaches a student: when a lecturer saves it, on the server, and again when the catalog arrives on the phone.

7. **Invariants survive edits.** "A completed photo step has a photo" holds after the fact as well. A completed step's photo can be retaken but not removed, and submitted records refuse every edit, in the store and (for deletes) on the server.

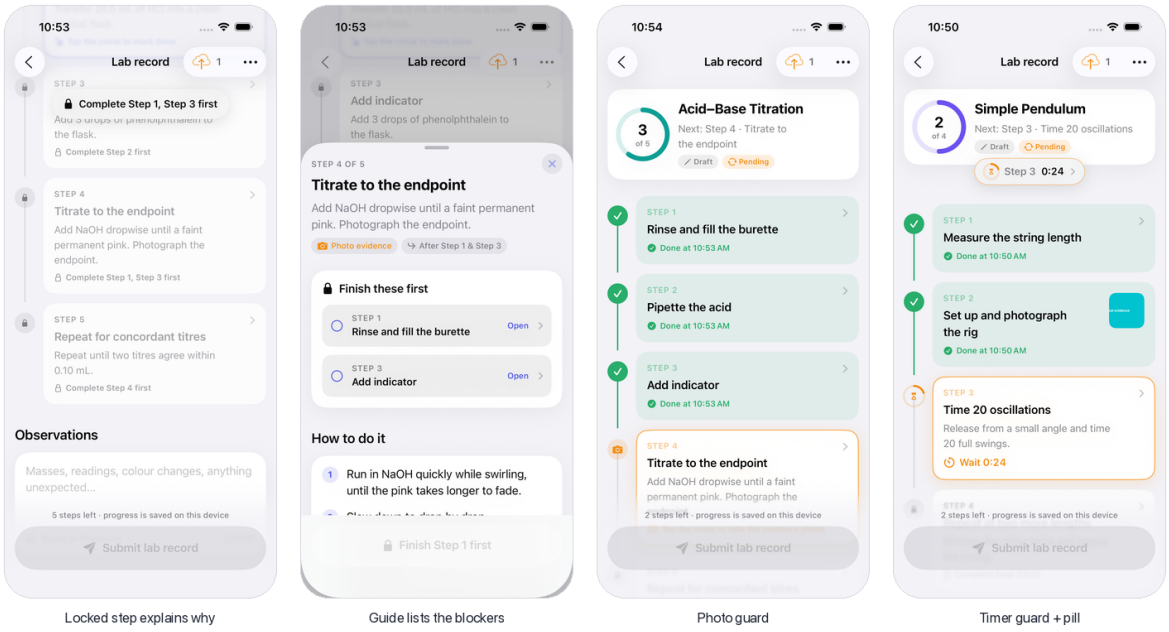


Figure 4. A locked step explains itself; the guide lists blocking steps; photo guard; timer guard with a live countdown.

5.5 Designing for a constrained device

The design also considers network capacity, battery use, storage and the student's attention. The table below explains the choices made for each constraint. Battery use and bandwidth were not measured in a controlled benchmark.

Constraint	Design response
Unreliable network	Local-first writes, outbox, batching (≤ 100), per-item results, backoff instead of tight retries
Bandwidth	Photos downscaled to 1600 px and JPEG 0.7 (about 100 KB); uploaded directly to Storage, never through the API
Battery	No polling: sync is driven by events (NWPathMonitor, local changes); the timer view only ticks while a timer runs; the Live Activity countdown is drawn by the system with no app

Constraint	Design response
	updates; GPS is a one-shot fix at 100 m accuracy with a 10 s timeout
Storage and memory	Evidence kept by relative file name; thumbnails generated with preparingThumbnail, not full-size images
Privacy and permissions	Camera, location and notification permission are asked only when the feature is first used (submit, photo, first timer), never at launch
User attention	One primary action per screen; capture mode hides the tab bar; Reduce Motion is respected

Tests: 13 constraint-engine unit tests. They cover multi-parent prerequisites, guard order, timer anchoring (including a first step's wait running from the record's start), messages, reachability, topological order, cycle detection, dangling and self edges, and the validity of every bundled experiment.

6. Native Hardware and Platform Integration

Capability	Implementation	Purpose	If unavailable
Camera	UIImagePickerController wrapped in CameraPicker; images saved by EvidenceStore	Photo evidence attached to a specific step; a photo automatically completes a photo-guarded step when the step is otherwise unlocked	Falls back to the photo library (Simulator); permission text set in NSCameraUsageDescription
GPS	LocationService (CoreLocation): one-shot requestLocation, 100 m accuracy, 10 s timeout, async continuation	Location stamp on submission, shown on a MapKit receipt	The record still submits with no location ("Not available")
Haptics	SwiftUI sensoryFeedback	Success when a step completes, warning when a locked step is tapped, a level change when a timer ends, success on submit	No effect
Live Activities / Dynamic Island	Widget extension with ActivityKit [13]; Text(timerInterval:) and ProgressView(timerInterval:)	The countdown continues outside the app; *Mark done* runs a LiveActivityIntent in the app process, still checked by the engine	Skipped if Live Activities are disabled
Notifications	UNUserNotificationCenter, time-interval trigger	"Step 3 is ready" when the wait ends; tapping it deep-links to the step guide	Skipped if permission is denied



Dynamic Island (compact) | Dynamic Island (expanded) | Lock Screen Live Activity

Figure 5. The compact Dynamic Island (ring and countdown); expanded (step, progress, ready time, Guide and Mark done); the Lock Screen Live Activity with the "ready" notification.

7. User Interface and User Experience

7.1 Navigation

Authentication is the entry point to the application (ADR-027). Student navigation comprises Labs, Records and Sync; lecturer navigation comprises Review, Labs and Sync (ADR-029). The student workflow progresses from a laboratory to an experiment briefing and then to its record. During capture, the tab bar is hidden and submission is available through a persistent bottom action. Lecturer navigation separates authoring and review from student participation.

7.2 Screen by screen

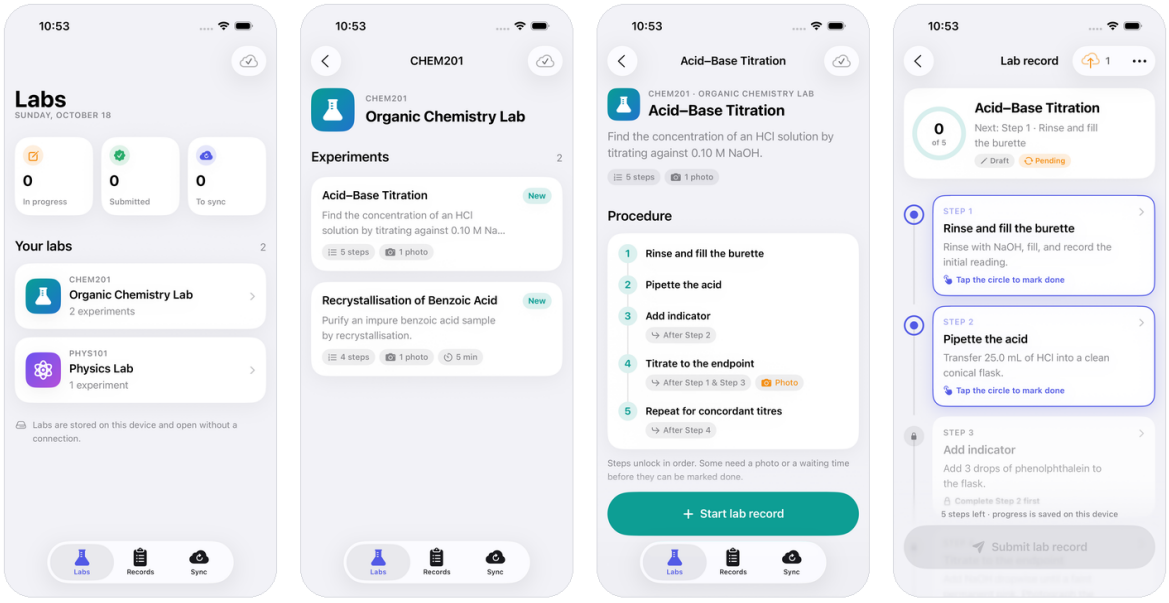


Figure 6. The Labs dashboard (date, stats, lab cards); a lab's experiments with their demands (steps, photos, waits) and progress; the experiment briefing with the procedure and its constraints, plus a sticky "Start lab record" button; a new record.

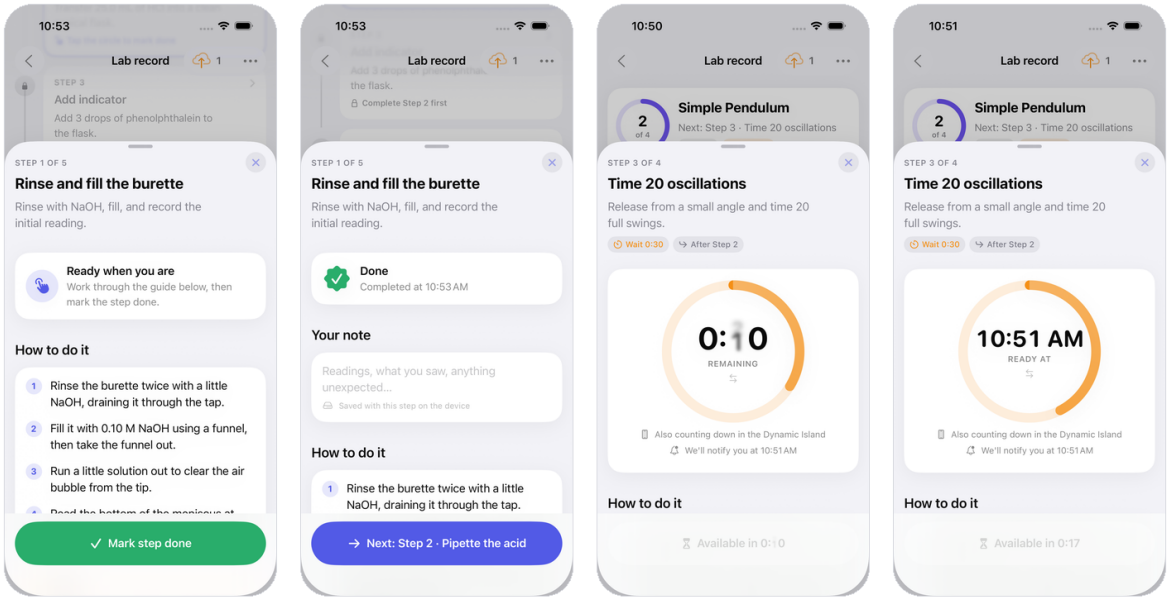


Figure 7. The step guide (how-to, safety, expected result, one primary action); after completion it offers the next step; the timer dial (tap to switch between remaining time and ready-at time).

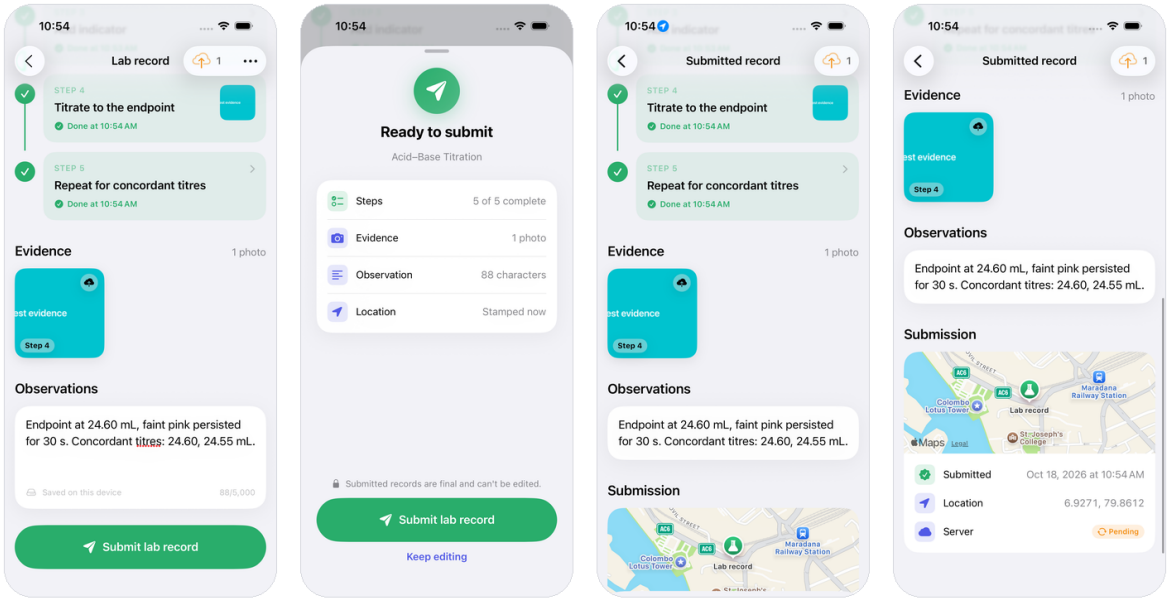


Figure 8. All steps done; the review-and-submit sheet summarising what will be handed in; the animated "Submitted" state; the read-only receipt with a GPS map.

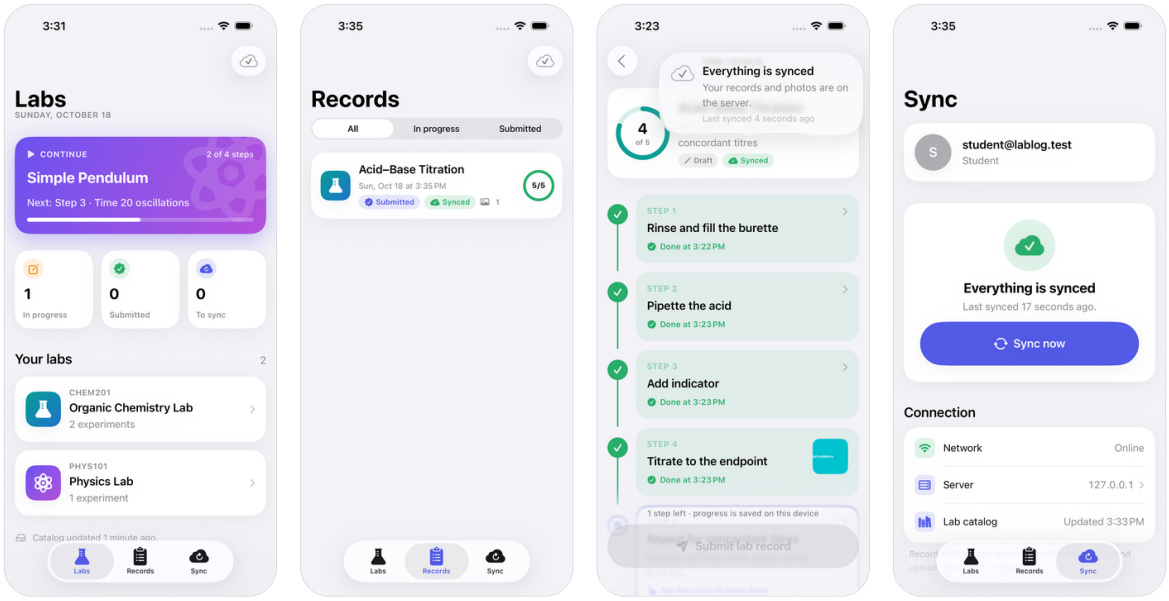


Figure 9. Home with a "Continue" card for the current draft; Records with filters, progress rings and sync badges; the toolbar's sync icon, which explains itself in one sentence when tapped and offers the action that helps (here everything is synced, so it only says when; signed out with work waiting, it offers *Sign in*); the Sync screen: the account, the sync status with *Sync now*, and the connection.

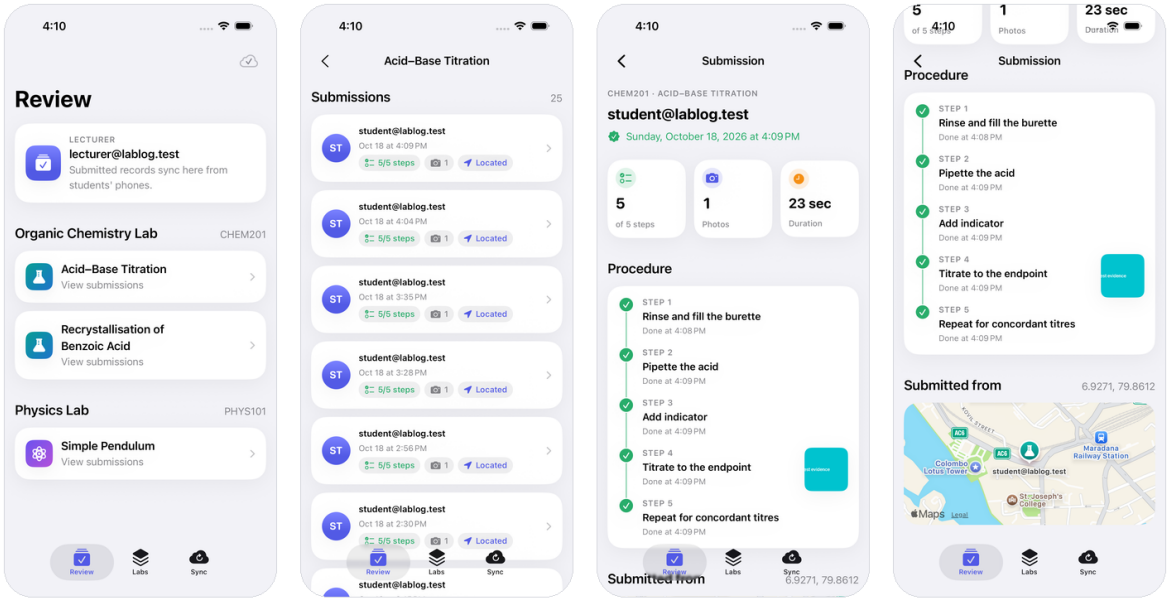
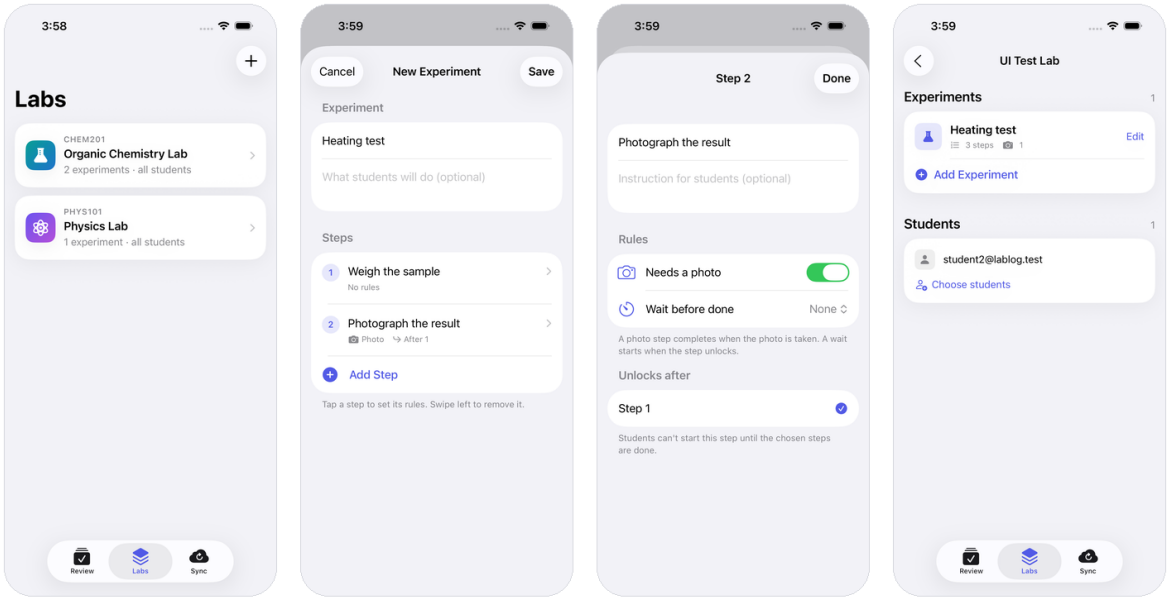
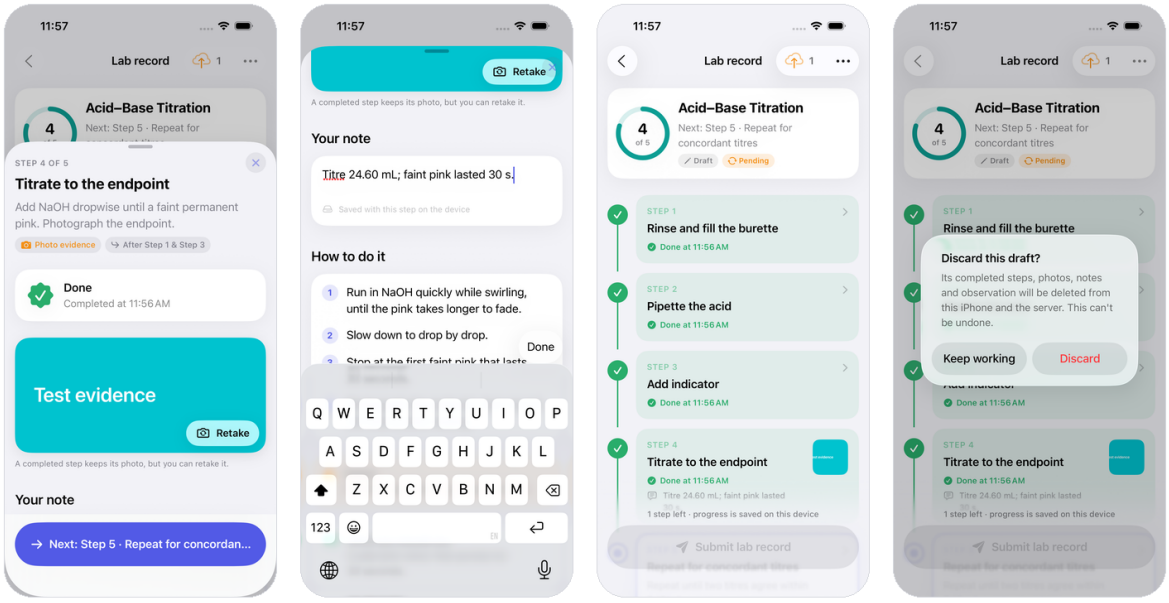


Figure 10. Signed in as a lecturer: the Review tab; an experiment's submissions; a submission's detail (steps done, photos, time taken, the step timeline with completion times and evidence); and where it was submitted. Lecturers have no Records tab: their tabs are Review, Labs and Sync (ADR-029).



Labs (lecturer) | New experiment | A step's rules | Experiments and students

Figure 11. A lecturer's Labs tab: the course labs (open to all students) and their own; a new experiment, each step summarised by its rules; a step's rules (needs a photo, a wait, and the earlier steps it unlocks after, which the constraint engine checks before saving); and a lab with its experiment (3 steps after an edit) and the one student enrolled in it (ADR-029).



Retake on the photo | A note on the step | The note in the checklist | Discard the draft

Figure 12. Retake sits on the photo itself (a completed step keeps a photo, so there is no Remove); a note on the step, with the action bar out of the keyboard's way; the note under its step in the checklist; discarding the draft, with a confirmation that says what will be deleted.

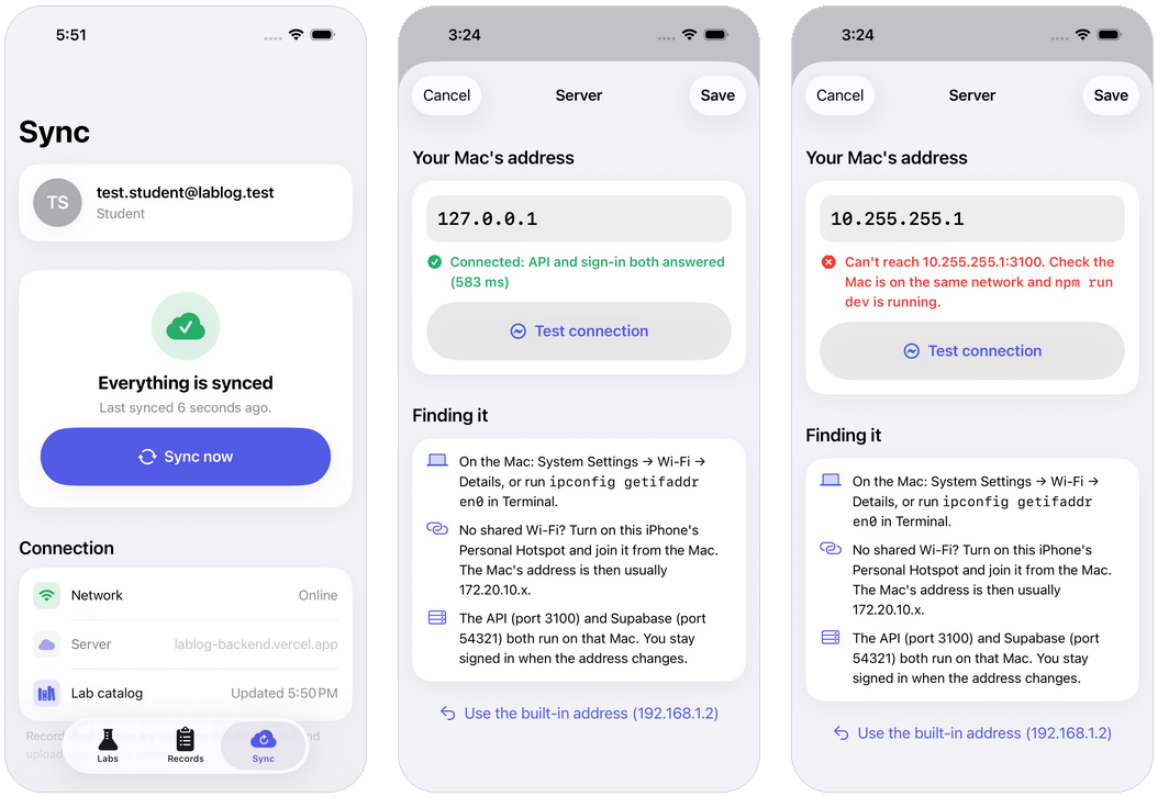


Figure 13. Server settings. The hosted build used on the iPhone and in the tests shows the fixed hosted address in Sync, and it cannot be edited (left). A local development build (ADR-021) can point at the Mac instead: the Server sheet tests an address against the API and sign-in before saving (centre), and an address that does not answer says what to check (right).

7.3 Interface design rationale

The interface makes the next available step easy to identify. Available steps stand out, completed rows become more compact, and locked steps explain what is missing. Selecting a locked step shows its conditions and links to the prerequisites. These choices follow Apple's Human Interface Guidelines [12] and keep the student's attention on the next action.

The recording screens use large tap targets, limited typing and immediate feedback. Taking a photograph can satisfy an evidence requirement, while haptics confirm a completed or blocked action. Before submission, students can review the contents of their record. The sync indicator separately shows whether work is saved locally or waiting to upload.

The app uses system text styles, semantic colours, platform symbols, VoiceOver descriptions and reduced-motion support. Dark appearance was checked through screenshots taken during testing.

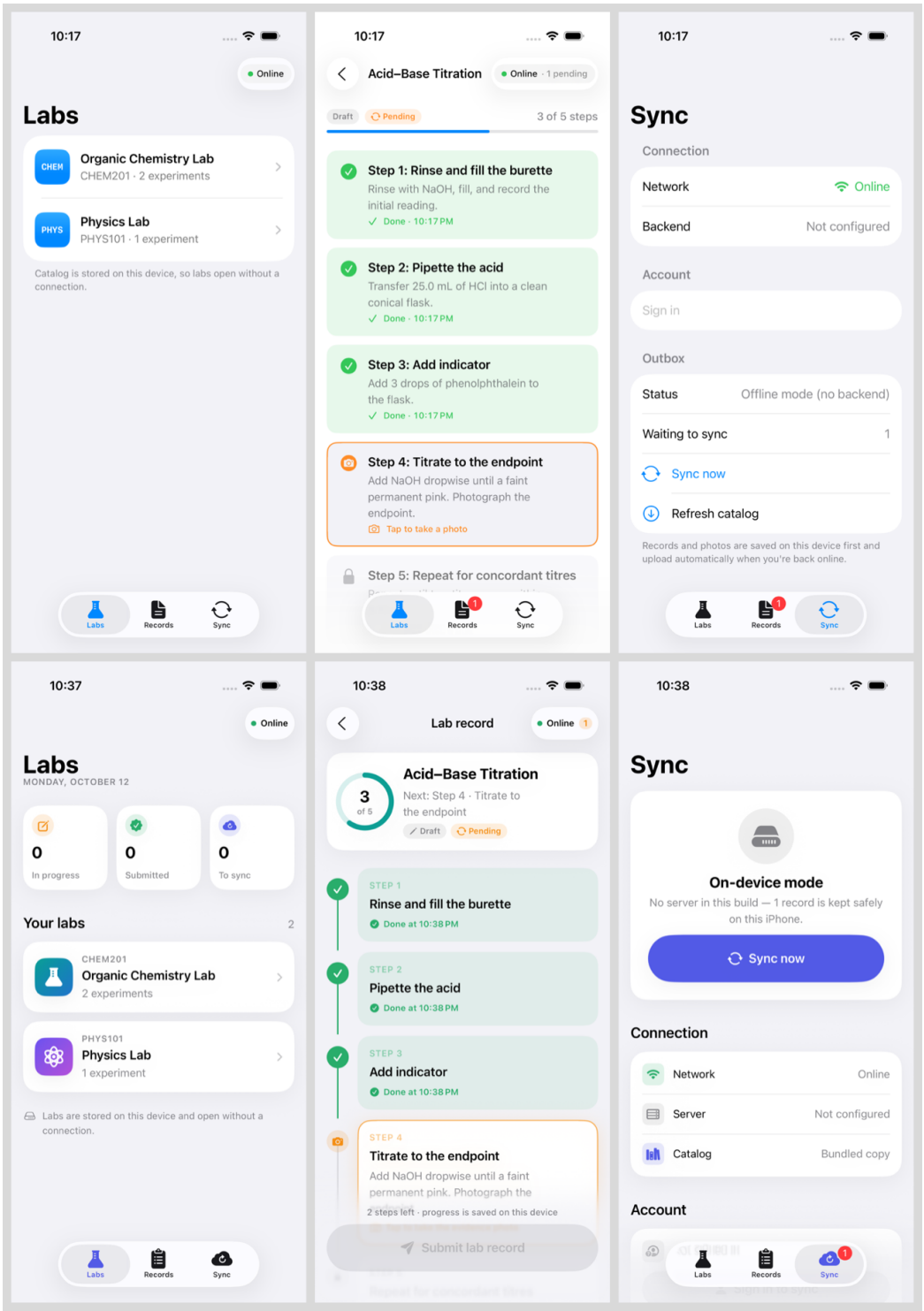
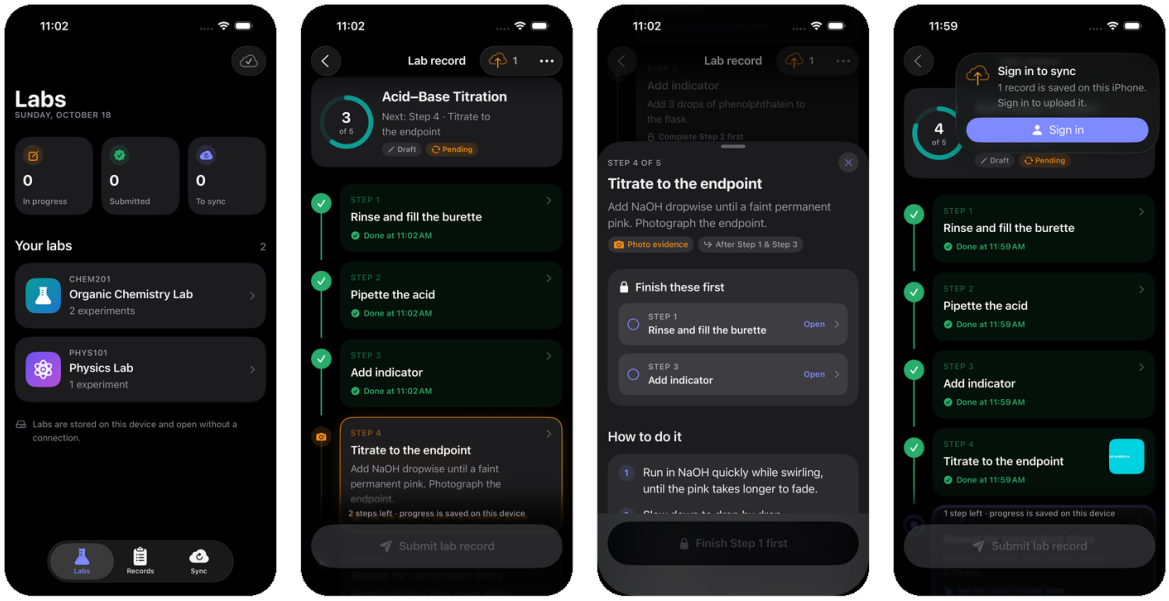


Figure 14. The first working build (top) used stock lists. The redesign (bottom) introduced the design system and the timeline checklist (ADR-013).



Labs | Checklist | Step guide | Sync icon explained

Figure 15. The same screens in dark mode. The app follows the system appearance: colours are semantic (system backgrounds and label colours), white text only sits on coloured fills, and card shadows are dropped in the dark. This was checked by running the UI tests with the simulator in dark appearance (16 Oct), and again after the sync icon replaced the Online pill (18 Oct, last panel). The first run produced 29 screenshots (prompt 19) and the second 30, as it added the sync icon's popover; all were reviewed for contrast, with nothing to fix.

8. Custom Component Design

8.1 ExperimentChecklist (the main custom component)

ExperimentChecklist is a reusable SwiftUI view that displays experimental steps as a vertical timeline. The parent supplies the steps and record state, while an injected StepConstraintEngine checks the rules. The checklist handles presentation and interaction; the parent model remains responsible for saving changes.

Properties (inputs)

Property	Type	Meaning
steps	[Step]	The steps to render (shown in order)
completions	Binding<[StepCompletion]>	Two-way bound completion state
recordId	UUID	The record that new completions belong to
startedAt	Date?	When the record began: a timed step with no prerequisites counts its wait from here (failure 028)
photoStepIds	Set<UUID>	Steps that already have a photo
photoThumbnails	[UUID: UIImage]	Optional evidence thumbnails shown on their step
isEditable	Bool	false for a submitted record (read-only)
engine	StepConstraintEngine	The rules (injectable, so it can be swapped in tests)
style	ChecklistStyle	Theme: spacing, radius, node size, tints, collapse behaviour

Events (outputs)

Event	Fires when
onStepCompleted(Step, StepCompletion)	A step was completed; the parent persists it, and successors re-evaluate automatically
onPhotoRequested(Step)	The circle of a photo-guarded step was tapped; the parent opens the camera

Event	Fires when
onLockedStepTapped(Step, String)	A locked step was tapped; the reason text is passed for display
onStepSelected(Step)?	A step card was tapped; the parent opens the step guide. If nil, a card tap behaves like a circle tap

Step notes are included in the completions binding and displayed by ChecklistRow. The row accepts a note property, so the checklist does not need another input for information already held in the completion.

Row states: *locked* (with reason), *available*, *needs photo*, *timer pending* (live ring and countdown), *completed* (with time and thumbnail).

How it works:

1. Each render builds a constraint context from completions, photoStepIds, startedAt and the current time, and maps every step to a row state through the engine.
2. A TimelineView re-renders once a second, but only while a timer guard is counting down.
3. When a countdown ends, the component fires a haptic.
4. Tapping a node runs the quick action (complete, request a photo, or explain the lock); tapping a card selects the step.
5. Completing a step appends to the binding and raises onStepCompleted, and the next render unlocks the successors.

Each row provides VoiceOver labels, values and hints. The timeline node and step card are separate accessibility elements because they perform different actions. This supports screen-reader use, but a full accessibility audit has not been completed.

```
ExperimentChecklist(
   steps: model.steps,
   completions: $model.completions,
   recordId: model.recordId,
   startedAt: model.record?.createdAt,
   photoStepIds: model.photoStepIds,
   photoThumbnails: thumbnails,
   isEditable: model.isEditable,
   onStepCompleted: { _, completion in model.persist(completion) },
   onPhotoRequested: { step in requestPhoto(for: step) },
   onLockedStepTapped: { _, reason in show(reason) },
   onStepSelected: { step in guide = GuideTarget(id: step.id) }
)
```

8.2 Other custom components

Component	Purpose
ChecklistRow	One timeline row: status node, connector, card; five visual states
StepGuideSheet	The guide: blockers you can jump to, interactive TimerDial, the single next action
TimerDial and LiveTimerPill	A countdown ring (tap to switch to ready-at time) and a floating live pill under the header
SubmitReviewSheet	Review → submitting → success, with an animated confirmation
SyncStatusButton	The toolbar sync icon: its look comes from a pure SyncIndicatorState, and a tap opens a one-sentence explanation with the action that helps (*Sync now* or *Sign in*)
ProgressRing, DisciplineIcon, StatTile, MetaChip, InfoRow, SyncStateBadge	Design-system building blocks
StepTimerLiveActivity	Dynamic Island (compact, minimal, expanded) and Lock Screen layouts
ReviewView, SubmissionsView, SubmissionDetailView	The lecturer's review flow: stats, step timeline, photos from signed URLs, location map
ManageLabsView, ExperimentEditorSheet, StepEditorSheet, StudentPickerSheet	The lecturer's lab management: lab cards, an editor for new and existing experiments whose step rows summarise their rules, a step's rules (photo, wait, unlocks after), and a student picker with search, select all and a clear selected state
SignInView	The first screen: the app icon, labelled fields, *Forgot password?*, and *Change server address* only when a local server cannot be reached

9. Decision records, failure logs and AI use

9.1 How the process was recorded

- **Decisions:** ADR-001 to ADR-030 in lablog-ios/docs/decision-records.md (summary in §3.2).

- **Failures:** lablog-ios/docs/failure-log.md, with the exact error text, an image, the cause and the fix. Raw logs sit next to each image in docs/screenshots/.

- **AI interactions:** development prompt records are provided in Appendix G, with what it produced and whether it worked.

9.2 Failure log

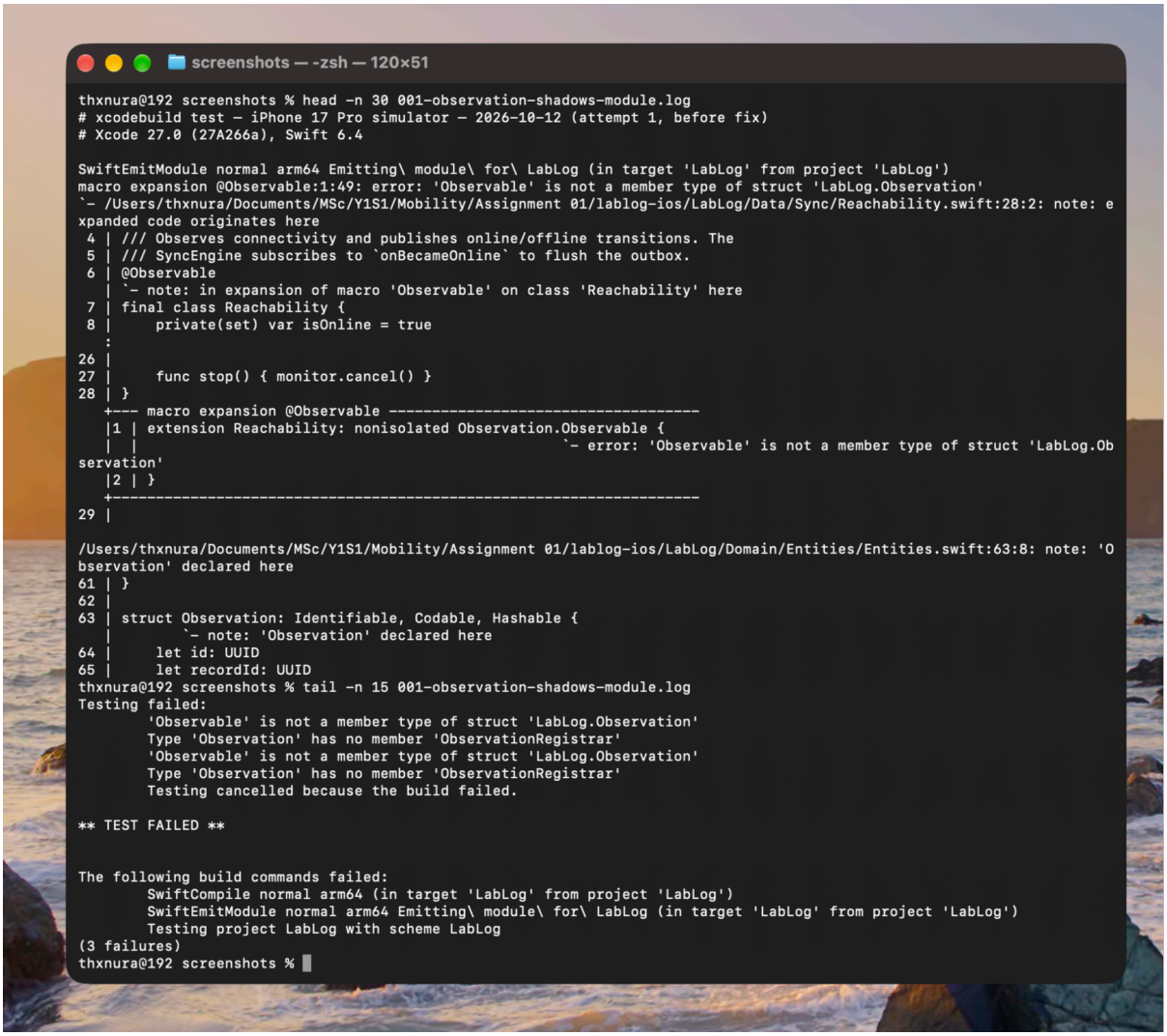
#	Date	Failure (exact error)	Cause	Fix
001	12 Oct	'Observable' is not a member type of struct 'LabLog.Observation'	My domain type Observation shadowed Apple's Observation module, which the @Observable macro expands into	Renamed it LabObservation
002	12 Oct	EXC_BREAKPOINT (SIGTRAP) in SwiftData; the test host crashed 16 times	The store kept the ModelContext but not its ModelContainer, which was freed	The store holds the container strongly
003	12 Oct	Badge wrapped as "Pend-ing", then showed only an icon; a row 4× too tall	SwiftUI Label adapts its layout to its container	Explicit HStack { Image; Text }
004	12 Oct	XCTAssertTrue failed — the UI test tapped the floating Submit bar	The test tapped an element's centre, which sat under the bar	Scroll into the clear band, then tap
005	12 Oct	Both UI tests: XCTAssertTrue failed; the Close button carried the id step-guide	A container's accessibility identifier overrode its children's	Identifiers on leaf elements only; combine before overlaying
006	12 Oct	Failed to tap "start-record" and a missing timer pill	A tap during a navigation push; the island press deep-linked into the guide	Wait for screens; handle the guide
007	12 Oct	The submit dialog appeared as a popover pointing at a step card	A confirmationDialog anchored to a large container	A custom Review & submit sheet
008	14 Oct	listen EADDRINUSE: address already in use :::3000	Another project's dev server held port 3000	The local API runs on 3100

#	Date	Failure (exact error)	Cause	Fix
009	14 Oct	Flaky sign-in test (three different errors)	Tab-bar overlap, sheet animation timing, keyboard focus	Auto-focus the email field; waits; hierarchy dump on failure
010	14 Oct	Receipt showed "Location: Not available"; then XCTAssertTrue failed twice	The restarted simulator had no simulated location (the app correctly submitted without one); a dropped tap during a transition; the test missed the 1.6 s success state	Tests pin the location (XCUIDevice.location), retry navigation taps, and watch for success and prompts together. GPS confirmed: "6.9271, 79.8612"
011	14 Oct	Lecturer UI test: tab tap ignored after launch; XCTAssertTrue failed (Submit disabled)	Cause of the ignored tap not found; Step 5 tapped while completed rows were re-laying out	Verified tab switching (title check, one retry); confirm each step completed
012	16 Oct	XCTAssertTrue failed waiting for step-note; the UI hierarchy showed Step 5's guide	The new Retake button sat under the sheet's floating "Next: Step 5" button, so the tap went to Next (the same class as 004, and a real UX flaw)	Retake (and Remove, offered only until the step is completed) float on the photo itself, so a completed step shows Retake alone (Figure 12); the test asserts Retake is hittable and the guide stays on Step 4
013	16 Oct	(Tooling) My keystrokes "ow" + Return landed in TermShot's Terminal window: zsh: command not found: ow	TermShot took the keyboard focus while I was typing, and its window ended in a live shell	It waits for keyboard and mouse to go idle, ends in a real shell with an empty keymap, and retakes the shot if a key was pressed. The faulty capture was overwritten, so there is no image
014	16 Oct	Failed to tap "Discard" Button: … Multiple matching elements found	iOS 26 exposes an alert's button to XCUITest twice	The test takes the first match inside the alert
015	16 Oct	Both sign-in UI tests: XCTAssertTrue failed - sign-in against Supabase Auth	The router moved the Mac from 192.168.1.8 to .88 mid-session; the app was built with the old address (the demo-day risk, for real)	Local UI tests then launched with -LabLogServerHost 127.0.0.1, and a local build on a phone used Sync → Server (ADR-021). Since ADR-025 and ADR-030 the phone and the tests use the hosted backend's fixed address, so this failure no longer applies

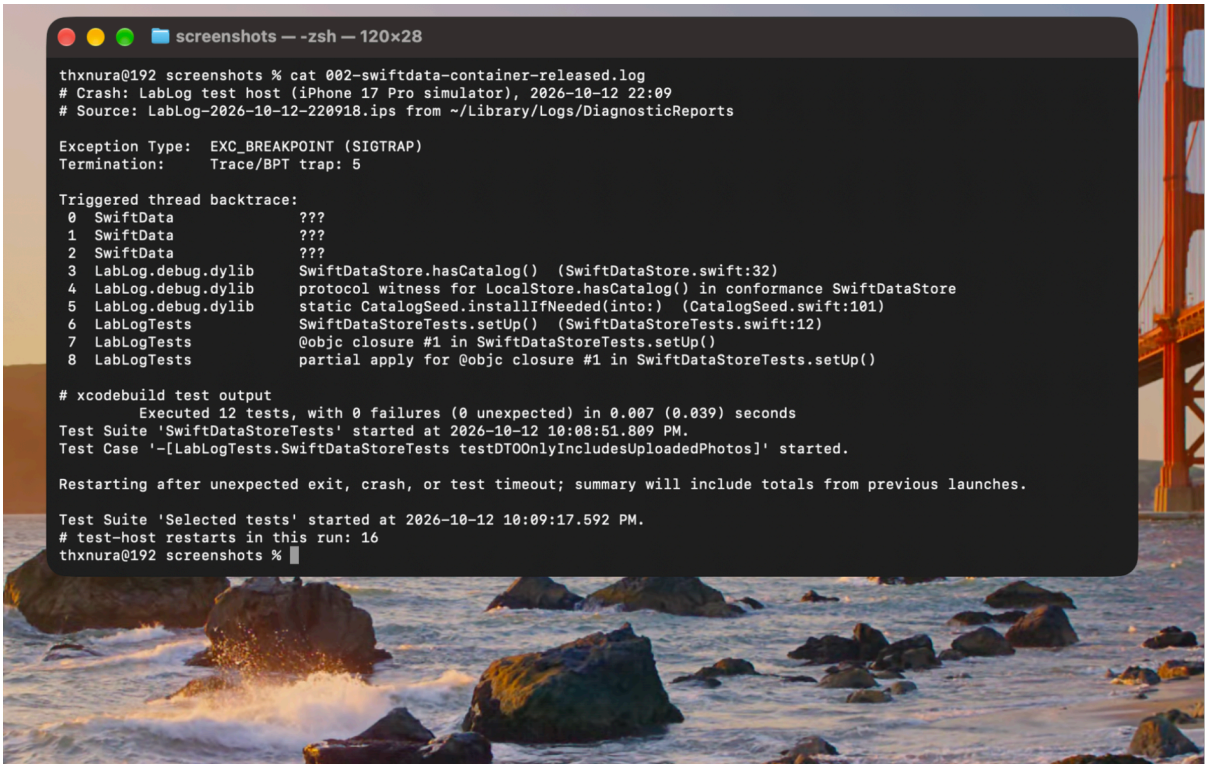
#	Date	Failure (exact error)	Cause	Fix
016	16 Oct	Full suite: titration XCTAssertTrue failed (Submit disabled) and sign-in XCTAssertTrue failed (no sheet)	A step circle tapped mid-layout never completed; one dropped tap	Every step is confirmed (completeStep); the sign-in tap retries once
017	18 Oct	UI tests failed in a different place each run: step-1 not found, start-record not found, Neither element nor any descendant has keyboard focus	The Mac was overloaded (load average near 28) and the simulator dropped taps and focus; the app itself was idle at 0% CPU	Test helpers confirm each tap, guide and keyboard and retry; the next full run passed. Four later dropped taps of the same kind (017b to 017e, the last a *Sign Out* tap) got the same treatment
018	18 Oct	The lecturer saw **0 Photos** on new submissions; Storage logs showed each photo uploaded, then DELETE 171 ms later; a new e2e test: + [] - ['ae7580fa-…']	The ADR-023 prune compared ids as strings: the app sends UUIDs in upper case, Postgres returns lower case, so the photo just sent looked "missing" and was deleted. Backend tests used lower-case ids, and the UI tests only checked "synced"	The API lower-cases every id when validating; a new e2e test sends upper-case ids; confirmed through the app (photo in Postgres and Storage)
019	18 Oct	failed - step-1 not found: "Start lab record" opened a "Submitted record" screen with 0, of 0 steps	Two SwiftUI tasks, one creating the screen's model and one loading it; when the load ran first nothing loaded, and an empty model reads as not-a-draft	One task creates and loads the model; the next full run passed
020	18 Oct	(UX, from my iPhone) The sync popover said "Sign in to sync" but tapping did nothing; then sign-in-popover not found in the new UI test	Sign-in was left to the Sync tab, so the one state that needs an action had none; the popover's container identifier replaced the new button's (as in 005)	A **Sign in** button opens the sign-in sheet in place; .accessibilityElement(children: .contain) on the container
021	18 Oct	(From my iPhone) Sign-in spun for 20 s, then a request timeout; curl to the old address: 000, to the new one: 200	The Mac's IP changed again (.88 → .2); the error didn't say what to do	Sign-in gives up after 8 s and says the server can't be reached, with **Change server address** in place; unit test; build default .2
022	18 Oct	sign-in-popover not found; the popover said "Everything is synced" in a test that never signs in	A lecturer session left in the Keychain by a test that failed before signing out	UI-test launches sign out, as they already clear timers and notifications

#	Date	Failure (exact error)	Cause	Fix
023	18 Oct	Lecturer UI test: Server returned 500: The connection to the database timed out (the app showed "Couldn't load submissions")	The local Supabase stack under load (Docker VM shared with another project, load average about 11)	None in code; passed on the rerun; another reason for hosting (§11)
024	18 Oct	(From my iPhone) After signing out, the previous student's records were still on the phone	Records were never tied to an account on the device; unsent ones would have uploaded under the next account	Sign-out clears the phone's records, warning about unsent work (ADR-026); unit test and UI check
025	18 Oct	All 5 UI tests: failed - signed in (sign-in against Supabase Auth) not found; backend tests: fetch failed; docker ps: daemon not running	Docker Desktop had stopped, taking the local Supabase stack down	Restarted Docker; the stack recovered and all tests passed
026	18 Oct	(From my iPhone) After signing back in, earlier submissions were missing	The app only pushed records, and sign-out now clears the phone	GET /api/records/mine and restoring after every sync, photos included (ADR-028)
027	18 Oct	Three UI tests: XCTAssertTrue failed, timer-dial not found	The shared test student had dozens of server records; after sign-in they were restored, so "Start lab record" reopened an old draft	Flow tests launch with -UITestSkipRestore (debug only); the sign-in test still checks the restore
028	18 Oct	(From my iPhone) A lecturer's timed first step never counted down; the Dynamic Island didn't match	With no prerequisite, the wait was measured from *now*, so every check restarted it	The wait runs from the record's start (engine, Live Activity, checklist startedAt); two unit tests
029	18 Oct	Lecturer UI test: Server returned 401: Invalid or expired token; later four UI tests: signed in … not found; docker ps: daemon not running	Docker Desktop quit three more times mid-run (15:04, 15:13, 15:54; its log shows an orderly quit each time, not a crash)	Restarted Docker; the hosted backend doesn't depend on it
030	18 Oct	XCTAssertTrue failed - the edit is saved (lecturer UI test)	The test read the edited experiment's row while the save was still in flight: the row already existed behind the sheet, so its wait ended at once (the screen recording shows Save spinning)	The test waits for the row's label to change; it passed, with the rest of the suite

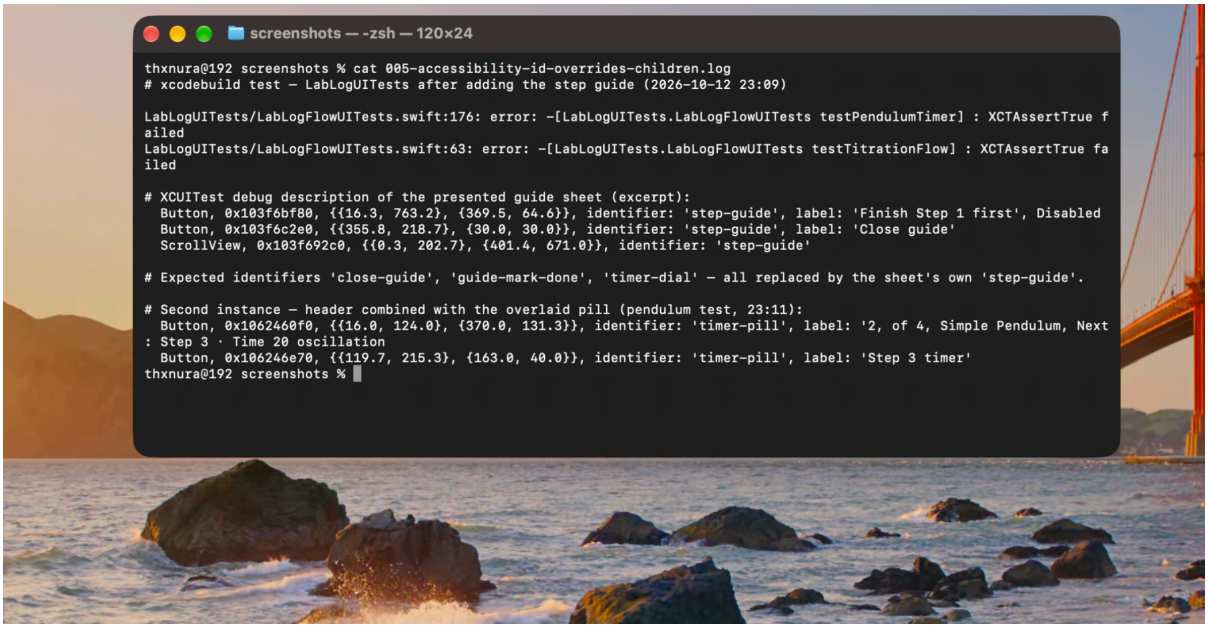
#	Date	Failure (exact error)	Cause	Fix
031	18 Oct	(Tooling) scripts/test-hosted.sh:30: read-only variable: status	In zsh, status is a built-in, read-only name for the last exit code, so assigning it stopped my new script before the tests ran	Renamed the variable; the next run passed
032	18 Oct	(On my iPhone) Failed to synthesize event: Neither element nor any descendant has keyboard focus, typing the password	The Passwords app's AutoFill bubble for my saved LabLog password covered the password field, so the test's tap hit the bubble and the email field kept the keyboard; the simulator has no saved passwords	The test presses Return to move to the password field and checks that this field has the keyboard
033	18 Oct	(On my iPhone) failed - experiment-Acid–Base Titration not found after signing in	iOS's "Save Password?" sheet sat over the app and took every tap until it was answered	The test taps *Not Now*; the next run on the iPhone passed all tests
034	18 Oct	(On my iPhone, real accounts) XCTAssertTrue failed - no submitted records left behind	Nothing had leaked: the second student opened on the Sync tab, the last tab the first student had used, because the starting tab was reset only when the role changed	The starting tab is chosen whenever the signed-in account changes; the rerun passed and the server was unchanged



Failure 001 — the first compile of the generated Xcode project.



Failure 002 — the crash report and test output for the SwiftData container lifetime bug.



Failure 005 — the debug description shows the Close button carrying its container's identifier.

Each error image is a Terminal window showing the log saved when the failure happened (excerpts of the xcodebuild output and crash reports, with my notes in # lines). The .log files sit beside the images in the repository. Five images (029b, 030b and, from the physical iPhone, 032b to 034b) are frames from the screen recording XCUITest keeps of each UI test. Failures

024, 026 and 028 were reported in words from my iPhone, so they have no image; each is covered by the tests named in its fix.

9.3 AI-Assisted Development

Claude Code assisted with code and test generation, debugging, interface refinement and documentation. I defined the requirements, made product decisions, reviewed suggested changes and tested the app on my iPhone. Automated tests and device checks identified issues that needed correction. Section 9.2 records the failures and fixes; Appendix G provides the original development prompts, outputs and outcomes.

Commit attribution. I wanted commits under my own name; AI use is fully disclosed in Section 9 and Appendix G.

10. Testing and Evaluation

Suite	Count	What it proves
StepConstraintEngineTests	13	Precedence, guard order, timer anchoring (including a first step's wait running from the record's start), messages, reachability, Kahn order, cycles, dangling/self edges, bundled catalog validity
SwiftDataStoreTests	18	Outbox rules, revision guard, duplicate completions, DTO contents, submit once, catalog replace, tombstones, photo removal, step notes, submitted records locked, sign-out clearing every record but keeping the catalog, restoring server records once without touching local ones, keeping only the labs the server lists
SyncEngineTests	9	Draining, photo-first ordering, per-item failure, network error, offline/signed-out, no backend, cyclic catalog skipped, a synced tombstone leaving the device, restoring records with their notes, observation and photos (the local copy wins)
ServerEndpointTests	5	Host replacement keeps scheme, ports and path; switching and resetting; host validation; sign-in to an address where nothing answers fails fast and says what to fix; a hosted backend has no address to change
ExperimentDraftTests	4	A lecturer's new experiment: the request carries rules by step number, removing a step renumbers the rules, editing an existing experiment keeps its step ids, and problems (including a loop found by the constraint engine) block saving
SyncIndicatorStateTests	4	The toolbar sync icon: quiet when synced, *Offline* wins over everything, waiting / failed /

Suite	Count	What it proves
		signed-out states, the on-device build
LiveActivityAndGuideTests	10	Which step drives the timer (a timed first step runs from the record's start), deep-link parsing, guides on every step, catalog upgrade, server JSON decoding, lecturer role read from the JWT (and not from user-editable metadata)
UI tests (XCUITest)	6	The full titration flow to submission (GPS stamp checked); timer, dial and Dynamic Island; sign-in and sync to the real backend; a student submits and a lecturer reviews it; retake, note, discard and the server-address check. Titration, timer and editing also run in dark appearance; a lecturer creates a lab, adds an experiment with step rules, edits it to add a step, and enrols a student. Every test signs in first, and the sign-in test signs out and back in to check the record is restored
Backend end-to-end (node --test)	20 (19 against the deployed API; the LAN-address test is local-only)	Auth (401), catalog and guides, idempotent sync, 422 validation, anti-hijack 403/409, signed upload retry, lecturer 403/200 with working photo URLs, storage URLs following the caller's host, retake replacing the stored photo, upper-case ids from the app keeping their photo, draft tombstones (idempotent, drafts only, owner only), a student getting their own records back (and nobody else's), lecturers creating labs with enrolment deciding who sees them, editing an experiment keeping its step ids (409 when a removed step has student work), a step loop refused with 422

The final runs on 18 October 2026 used the hosted backend (ADR-030). All 63 unit tests and six UI tests passed on the iPhone 17 Pro simulator and again on a physical iPhone 17 Pro. The

first runs on the phone exposed two device-only test problems caused by the iOS Passwords app (failures 032 and 033), which were fixed in the tests. The backend suite passed against the deployed API with 19 passes and one skip, for a LAN-address test that applies only to a local stack; an earlier local run passed all 20. A read-only check on the phone with the real demo accounts confirmed that signing in restored the student's 11 records with their photos, and that the next student to sign in saw none of them; it also found that the next student opened on the previous student's last tab (failure 034, fixed). Test runs use isolated test accounts whose data is deleted afterwards [2] to [5].

11. Limitations and Future Work

11.1 Operational and synchronisation limits

Hosting the API and database removed the need to keep the development laptop at a fixed network address. ADR-030 describes the current hosted testing process, while local Supabase remains available for development. The earlier network-address and Docker failures are retained in the failure log. Availability, recovery time and sustained-load capacity have not been measured.

Synchronisation assumes that each record has one author, with the server accepting the last write. The local revision check protects edits made during an upload, but it does not merge changes from several devices. Restoring records adds server records that are missing locally and keeps existing local copies. Shared practicals and concurrent editing would need a separate conflict-resolution strategy.

11.2 Security and evidence limits

The backend checks authentication, ownership and payload structure, and restricts direct database access. However, the procedural rules are mainly enforced by the app. A modified client could bypass them unless the server also checks those rules when accepting a record. Sync also does not recheck laboratory enrolment for every record. Adding these server-side checks is a priority for future work.

Photographs, completion times and location help a lecturer review the record, but their authenticity is not independently verified. Students can submit when location is unavailable, and the system does not check what a photograph contains. The application therefore cannot be claimed to prevent fabricated records.

11.3 Functional scope and evaluation limits

Lecturers need a connection to review submissions and manage labs. Marks, lecturer feedback, deletion of labs and experiments, group records and Android support have not been implemented. Live Activities rely on platform timing for readiness updates; push updates could improve this behaviour. Layouts designed specifically for iPad are also future work.

The evaluation consists of functional tests and documented device checks. It does not measure accessibility conformance, battery use, sync throughput, long-term reliability or learning

outcomes. Further testing should involve students and lecturers, network interruptions across multiple devices, security checks and measurements of resource use.

12. Conclusion

LabLog provides a working native application for recording practical laboratory work in the assigned Edtech domain. Local storage allows students to continue recording offline, while a shared constraint engine checks step dependencies, waiting periods and evidence requirements. Camera capture, timed activities and lecturer workflows address the main SE5070 requirements. The repositories, decision records, failure log and original prompts explain how the project was developed [2] to [5].

The tests confirm these behaviours on the simulator, on a physical iPhone and against the deployed backend. The next priorities, from Section 11, are stronger server-side checks, clearer conflict handling and testing with intended users. The project shows how offline recording and procedural rules can be brought together in a mobile educational application.

References

[1] L. Petthawadu, *SE5070 Enterprise Mobility: Individual Assignment*, MSc in IT, Enterprise Application Development, Year 1, Semester 1, 2026. Supplied course brief: Assignment - Enterprise Mobility 2026-2.pdf.

[2] D. T. M. Peiris, *LabLog iOS Application: Source Code, Tests and Project Documentation*, 2026. Project repository: https://github.com/thxnura/lablog-ios. Local implementation supplied with this report.

[3] D. T. M. Peiris, *LabLog Backend: Source Code, Schema and End-to-End Tests*, 2026. Project repository: https://github.com/thxnura/lablog-backend. Local implementation supplied with this report.

[4] LabLog project, *Architecture Decision Records*, ADR-001 to ADR-030, 2026. Project artefact: lablog-ios/docs/decision-records.md.

[5] LabLog project, *Failure Log and AI Interaction Register*, 2026. Project artefact: lablog-ios/docs/failure-log.md; development summary in Section 9 and original prompt register in Appendix G.

[6] A. Hofstein and V. N. Lunetta, "The laboratory in science education: Foundations for the twenty-first century," *Science Education*, vol. 88, no. 1, pp. 28-54, 2004, doi: 10.1002/sce.10106.

[7] S. Kanza, C. Willoughby, N. Gibbins, R. Whitby, J. G. Frey, J. Erjavec, K. Zupančič, M. Hren and K. Kovač, "Electronic lab notebooks: can they replace paper?," *Journal of Cheminformatics*, vol. 9, art. 31, 2017, doi: 10.1186/s13321-017-0221-3.

[8] M. Kleppmann, A. Wiggins, P. van Hardenberg and M. McGranaghan, "Local-first software: You own your data, in spite of the cloud," in *Proc. ACM SIGPLAN Int. Symp. on New Ideas, New Paradigms, and Reflections on Programming and Software (Onward! '19)*, 2019, pp. 154-178, doi: 10.1145/3359591.3359737.

[9] A. B. Kahn, "Topological sorting of large networks," *Communications of the ACM*, vol. 5, no. 11, pp. 558-562, 1962, doi: 10.1145/368996.369025.

[10] M. Fowler, *Patterns of Enterprise Application Architecture*. Boston, MA, USA: Addison-Wesley, 2002.

[11] C. Richardson, "Pattern: Transactional outbox," microservices.io. https://microservices.io/patterns/data/transactional-outbox.html (accessed 18 Oct. 2026).

[12] Apple Inc., "Human Interface Guidelines," Apple Developer. https://developer.apple.com/design/human-interface-guidelines/ (accessed 18 Oct. 2026).

[13] Apple Inc., "ActivityKit," Apple Developer Documentation. https://developer.apple.com/documentation/activitykit (accessed 18 Oct. 2026).

[14] Supabase, "Row Level Security," Supabase Docs. https://supabase.com/docs/guides/database/postgres/row-level-security (accessed 18 Oct. 2026).

Appendices

A. Running the project

The hosted backend needs no local services (ADR-030). The deployed API is at https://lablog-backend.vercel.app/api.

```
# Backend: the 20 end-to-end tests against the deployed API (needs .env.hosted)
cd lablog-backend && npm install
npm run test:hosted

# iOS: with Config/Secrets.hosted.xcconfig, every build uses the hosted backend;
# 63 unit + 6 UI tests, signed in as test-only accounts, cleaned up afterwards
cd lablog-ios && scripts/test-hosted.sh
```

Optional local stack for offline development (Docker and the Supabase CLI): supabase start, npm run local:setup -- --host <Mac LAN IP>, npm run dev and npm run test:e2e, with the printed values in Config/Secrets.xcconfig and no hosted xcconfig. Its demo accounts are student@lablog.test / LabLog-Student-1 and lecturer@lablog.test / LabLog-Lecturer-1; the hosted accounts have private passwords. A build with neither config has no backend and therefore no accounts: it opens directly on Labs in on-device mode, with the bundled catalogue, and records stay on the device. That is the only case without sign-in; every build with a backend, including the submitted one, signs in first (ADR-027).

B. Environment

Xcode 27.0 · Swift 6.4 · deployment target iOS 17 · tested on an iPhone 17 Pro (iOS 27.2) and the iPhone 17 Pro simulator (iOS 26.0.1) · Node 22 · Express 4 · zod 3 · Supabase CLI 2.53 · Postgres 17.

C. Size

About 8,400 lines of Swift in the app and widget, 1,500 lines of Swift tests, 1,300 lines of TypeScript (API and scripts), and 740 lines of SQL and end-to-end tests.

D. Academic integrity

The project declaration states that illustrations and diagrams were created in code or captured from the app, and that no existing application codebase was copied. The frameworks, libraries and platform symbols used are listed below. The custom checklist and constraint logic were developed within this project, with AI assistance disclosed in Section 9.

Area	Libraries, frameworks and tools used
iOS app (Apple frameworks)	SwiftUI, UIKit (camera picker), SwiftData, Observation, CoreLocation, MapKit, Network (NWPathMonitor), Security (Keychain), UserNotifications, OSLog
Widget extension	WidgetKit, ActivityKit (Live Activity / Dynamic Island), AppIntents
iOS tests	XCTest, XCUITest
Backend runtime	Express, zod, @supabase/supabase-js, cors, dotenv
Backend tooling	TypeScript, tsx, Node 22 test runner
Infrastructure	Supabase (Postgres, Auth, Storage), hosted and via the Supabase CLI; Vercel (the API); Docker (optional local stack)

Help to or from batch mates: None. No help was given to or received from batch mates.

E. Assignment Requirements Traceability

Assignment requirement [1]	Report evidence
Index digit sum and project area	Section 1
Project idea and feature overview	Section 2
Architecture, decisions, data flow and source organisation	Section 3; ADR register [4]
Offline data access and synchronisation	Section 4; evaluation in Section 10
Constraint algorithms and principles	Section 5
Native hardware integration	Section 6
UI/UX flow with screenshots	Section 7
Custom component properties and events	Section 8
Decision records, failures and AI prompts	Section 9; Appendix G; project logs [4], [5]
Source code links	Cover details; references [2], [3]

F. Report Preparation Note

The supplied Enterprise Software Architecture Assignment 2 report was used as a reference for the document layout. Its financial-services content was not used as evidence for LabLog.

AI assistance was used to edit the prose and prepare the Word document. Original prompts in Appendix G and diagnostic records in Section 9 have been preserved.

G. AI-Assisted Development: Development Prompt Register

This register covers development prompts 1 to 48, retaining their original wording, outputs and outcomes. Later interactions are recorded in lablog-ios/docs/failure-log.md. Section 9.3 explains how the assistance was used and reviewed.

#	Exact prompt	What it produced	Result
1	"check the status of this project and install the app in my iphone"	Status review, the Xcode project, signing, and installation on my iPhone	Partially (failure 001)
2	"it shows scaffold ready, wire lab view and other stuff, build the full app, and if u detect errors, those should be screenshoted and make a doc"	The full app (store, sync, auth, screens, camera/GPS wiring, tests) and a build log	Partially (failure 002; a race in its record screen surfaced on 18 Oct as failure 019)
3	"i get this error repetatively" (with a screenshot of the crash dialog)	A diagnosis from the crash report and the fix	Worked
4	"its look too much basic, make ui ux more like other porffesional app design with apple design skills"	The design system and a redesign of every screen	Partially (failures 003, 004)
5	"yes and u shoeud authorize as me, do not include u"	Commits under my own identity	Worked
6	"add more interactibe animations like when timer goes show them in dynamic island and show interactive timer, and when we click step add step guide as well"	The Live Activity, timer dial, step guides and deep links	Partially (failures 005, 006)
7	"yes rewrite that commits and remove all clade coauth and add my photo and name"	A history rewrite with authorship preserved	Worked
8	"this submit dialog look ugly, fix it, also make dynamic island more good with details and make proffesional"	The review sheet and a richer Live Activity	Worked (fixed failure 007)
9	"yes continue and do next build"	The local Supabase stack, end-to-end tests, lecturer role, ownership checks, and real sync from the app	Partially (failures 008, 009)
10	"what more we have to build"	A gap analysis against this brief	Worked

#	Exact prompt	What it produced	Result
11	"ok do next ones"	This report draft, its diagrams and figures, demo notes, and the in-app lecturer review; checking the figures exposed failure 010	Partially (failures 010, 011); report is a first draft for my review
12	"check what we more have to build"	A second gap check: submission tasks, the risk of a LAN-only backend at a demo, optional small builds	Worked
13	"ask me the points should be filled in Appendix D"	Questions on batch mates, assets and libraries; Appendix D filled from my answers	Worked
14	"termiansl erros all looks like styles neh, for this lets do this, make small mac app, not inside this repo. make it inside documemnt/thanura and when we fiving logs, inte rendered same to mac terminal and we can capture ss from it", then "look its huge different", "now this is too mcuh perfect, adding shadow and all…", "i said u to take ss like this" (with my own screenshot)	TermShot, a small Mac app that shows a saved log in a real Terminal window on my desktop and saves the screenshot; the error images were re-taken with it in the style of my own screenshot	Partially (window title, margin marks, inactive windows, a captured keystroke (failure 013) and two style rounds, all fixed)
15	"make word document"	LabLog-Report.docx, generated from the same Markdown (headings, tables, code, 14 images, page numbers)	Worked after two conversion bugs were fixed (lost spaces between links, a skipped image)
16	"so whats next we have to build"	A plan: a runtime server address first (the demo network risk), then discard draft, photo retake and step notes	Worked
17	"ok do next"	The four features above, across app and API, with 10 unit tests, 4 backend tests and a UI test; ADR-021 to 023	Partially (failure 012 caught by the new UI test and fixed; the photo prune it added also deleted every new photo, found two days later as failure 018)
18	"so whats next"	A short plan: update my demo notes, check dark mode, then stop adding features before the deadline	Worked
19	"so update md and do the checks"	My demo notes updated for the current app (a Codable trap caught and verified along the way); UI tests run in dark mode, 29 screens reviewed, Figure 15	Worked (no fixes needed)

#	Exact prompt	What it produced	Result
20	"i checked the readme and other stuff, i cant give this repos to sir, make it ready to submit"	Both repos prepared for a reader: new READMEs, the report PDF in the repo, the failure log reorganised (index, failures in order, prompts table), template notes and exam wording removed, a broken lint script dropped	Worked
21	"make repo public"	Full git history scanned for secrets (none), then both repos made public and checked without signing in	Worked
22	"btw in readme and dark mode ss"	The dark-mode figure added to the README	Worked
23	"btw make app more production ready, i mean this style online status look odd, make it production ready" (with a screenshot of the toolbar)	The "● Online 1" pill replaced by a quiet sync icon with an explanatory popover and Sync now (ADR-024), 4 unit tests, new screenshots	Partially (failure 017; the screenshot review found a signed-out icon that read as an account button, a notification left over from another test, "Your 0 records" wording and failure 018; the final full run found failure 019; all fixed. The popover still had no way to sign in: failure 020, reported from my iPhone)
24	"when i click here itd not prompt to signin, fix it" (with a screenshot of the sync popover from my iPhone)	A **Sign in** button in the signed-out popover that opens the sign-in sheet in place; the action modelled in SyncIndicatorState with a unit test; the UI test taps it	Partially (failure 020: the new UI test step failed first because the popover's identifier hid the button's, as in 005; fixed)
25	"signin takes too much time, check whats happening", then "i mean request timeout"	The cause found (the Mac's IP changed again); sign-in now fails in 8 s with a clear message and **Change server address**; a unit test; the fixed build on my iPhone	Worked (failure 021; the full run that followed found failure 022, fixed)
26	"push the db to supabase now cli done"	The linked hosted project got the schema, the seed and a new migration that turns on RLS for every table and creates the private photo bucket (ADR-025); checked with the service role and the anon key	Worked
27	"now create test user"	The three test accounts on the hosted project (same emails and roles, new private passwords, since the	Worked

#	Exact prompt	What it produced	Result
		README's local passwords are public); each checked by signing in and reading its role from the token	
28	"ok um ready to deply this in vercel, show me env to add", then "open .env"	The API made ready for Vercel's zero-configuration Express (default export of src/app.ts, ES module), checked locally; the four variables in a private file opened in VS Code, the service-role key kept out of the chat	Worked
29	"ok now make vercel.json"	vercel.json running the function in Seoul (icn1), next to the database	Worked
30	"https://lablog-backend.vercel.app test this"	A smoke test of the deployed API, as the app uses it (15 checks), with its test record cleaned up	Worked after one fix to the test itself: it looked for guideSteps, but the API returns guide.steps
31	"now point it in app"	iPhone builds use the hosted backend through a git-ignored, iPhone-only config; the simulator and tests stay local; the Server row is read-only and the sign-in error fits a hosted build; 2 unit tests; installed on my iPhone	Worked
32	"it show invalid password" (signing in as the lecturer on my iPhone)	Checked that the hosted lecturer account signs in (200) and that the phone's attempt reached the hosted server, not the Mac; the likely causes: an old saved password filled in by iOS, or a typo	Worked (no fault found in the app)
33	"also when we logout it show offline data, fix it"	Sign-out clears the phone's records with a warning for unsent work, an open record closes itself, a unit test and a UI check (ADR-026)	Worked (failure 024)
34	"in assignmnt check is there any requirmen to keep guest login without signin", "check this" (the brief), "no i need to check are we need to keep guest login without signin"	The brief read in full: it never mentions login, accounts or guests; it requires a backend and offline work that syncs back, so either model meets it	Worked
35	"so dont allow them to use app without signin, keep offline sync as but when they logout do not allow, like whatsapp work"	The app opens on sign-in and can't be used signed out; offline after the first sign-in; sign-out returns to sign-in and clears the phone; a local build can change the server from the sign-in	Worked

#	Exact prompt	What it produced	Result
		screen; UI tests sign in at launch (ADR-027)	
36	"make login ui more good with apple design and make settngs page more good ui ux"	A first redesign: a welcome screen with a glowing brand mark and three feature rows, and a Settings-style Sync page (account card, sections, Sign Out at the bottom)	Partially: I said it looked too "vibe coded"
37	"its lokk too vibe coded, mak it simple", then "i need simple but proffesional one", then "only show the version, no need to show the server data in login"	Sign-in rebuilt plainly: the real app icon, "Sign in to LabLog", labelled Email and Password fields with show/hide, "Forgot password?" (accounts are reset by the university), one Sign In button, only the version at the bottom; the Sync page's account card as a plain monogram	Worked after the three rounds
38	"ok i found a bug, when i signin and check my prevous lab subsmition, there are missing , fix it"	GET /api/records/mine (backend, deployed), restoring records and their photos after sign-in (app), tests at every level, the fix on my iPhone (ADR-028)	Worked (failure 026; Docker had also stopped, failure 025)
39	"lecturers no need access labs neh, they will not do the labes neh, add option to create labs to lecturers , and poin who can do this" (with a screenshot of the lecturer's Records tab); my answers: the lecturer picks the students, full step rules	Lecturer tabs Review · Labs · Sync; create a lab, add experiments with photo, wait and prerequisite rules (checked by the constraint engine in the app and with Kahn's algorithm on the server), choose students; students see only open or enrolled labs (ADR-029); backend, unit and UI tests; deployed and installed	Worked (failure 027 on the way, from the restore feature)
40	"when we click save and create, its wait until backend respond, make it optimized"	Create answers with the new lab or experiment whole (server step ids) and the app adds it to its store and closes the sheet at once, with no catalog refetch; choosing students saves optimistically and reverts on failure; the catalog refresh fetches experiments concurrently instead of one by one	Worked
41	"make student choose ui more good"	The student picker rebuilt: a monogram, name and email per student, a checkmark circle that fills when chosen, "2 of 5 selected" with Select All, always-visible search, empty and no-results states, Save only when something changed	Worked

#	Exact prompt	What it produced	Result
42	"also we cant change exiting expermants steps fix it", then "and make ui better of them"	Editing experiments: PUT /api/manage/experiments/:id keeps step ids and refuses to remove a step with students' work (backend test); one editor for new and existing experiments; step rows with a numbered circle and rule symbols (camera, timer, after); experiment rows showing steps, photos and timers; unit test and the lecturer UI test now edits an experiment	Worked
43	"i created test lab and counter now not going check why, also dynamic island not working"	Found the engine's fallback to *now* for a wait with no prerequisites; the wait now runs from the record's start in the engine, the Live Activity and the checklist component; two unit tests; installed on my iPhone	Worked (failure 028; Docker had quit again during the tests, failure 029; checking the fix found failures 030 and 017e, both in the tests)
44	"is all done"	A status check of both repos: everything committed, the iOS repo 8 commits ahead of GitHub; the items only I can do (read and submit the report, reinstall before the signing expires, turn off public sign-up, Docker, viva practice)	Worked
45	"push as me"	The 8 iOS commits checked for my authorship and no co-author lines, then pushed	Worked
46	"did u update the doc ss and markdoen and all"	An audit of the shared doc, screenshots and Markdown. Fixed: four doc sections still showing old screens, the doc's summary, the iOS README ("sign in from the Sync tab", no lecturer figure), the screenshots README and the viva notes; README changes pushed	Worked (gaps found and fixed)
47	"update the report as well"	A full read of the report against the app. Updated: lecturer lab authoring in the idea, the read path, the API table, the constraint chapter (a wait anchored on the record's start; validation when an experiment is written), the checklist's startedAt, the lecturer components, the failure and test notes, my own role, limitations; PDF and Word rebuilt	Worked

#	Exact prompt	What it produced	Result
48	"ok what are the more gaps have, also do not use docker, us the vercel link for testog and make finalzie"	A gap check against the brief; every test moved to the hosted backend (ADR-030): test-only accounts, the backend suite against the deployed API, the simulator and UI tests on the hosted build, a clean-up before and after each run; docs, report and both repos finalised	Worked (failure 031, in the new script)